STAR: A Unified Spatiotemporal Fusion Framework for Satellite Video Object Tracking

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***Abstract*— Satellite video object tracking (SVOT) delivers comprehensive spatiotemporal insights for Earth surface observation, yet existing SVOT methods confront several critical challenges, including data scarcity, modality restrictions, paradigm gaps, and underutilization of multidimensional features, sealing the performance ceiling. This study proposes STAR, a unified spatiotemporal fusion framework for SVOT, mitigating these issues. To optimize satellite video (SV) scenes, STAR first introduces a scene enhancement module (SEM) for generating enhanced multimodal representations. Then, the extraction-correlation-adaptation module (ECAM) is designed, incorporating a multimodal hierarchical Transformer architecture with local and unified relation modeling, which jointly achieves feature extraction, relation learning, and domain adaptation. In addition, the temporal decoding structure is introduced to integrate deep temporal features via attention propagation. Finally, the inertial navigation module (INM) models physical temporal features, including an awareness selector to assess the tracking confidence-uncertainty and an inertial navigation scheme to manage anomalous interferences and continuous trajectory. Inspired by the prompt learning pattern, STAR introduces a minimal number of tunable parameters yet achieves competitive performance across various SVOT benchmarks. Implementation details and evaluation results are available at https://github.com/YZCU/STAR**

***Index Terms*— Multimodal tracking, object tracking, satellite video, spatiotemporal fusion.**

I. Introduction

SATELLITE video object tracking (SVOT) aims to accurately determine the state of an object in subsequent frames given its initial state, with applications spanning traffic analysis, national defense security, and ocean monitoring [1]. In contrast to generic video (GV) captured by common devices, SVOT presents unique characteristics. As shown in Fig. 1, due to the low resolution and long satellite shooting distance, objects occupy few pixels and exhibit limited spatial features (i.e., weak objects), making precise localization difficult [2]. In addition, satellite video (SV) is captured from high-speed satellite platforms. Small objects in nonstationary backgrounds are prone to anomalous interferences like background clutter (BC), occlusion, and dense similarity (DS) [3]. Achieving robust SVOT remains challenging due to several factors.

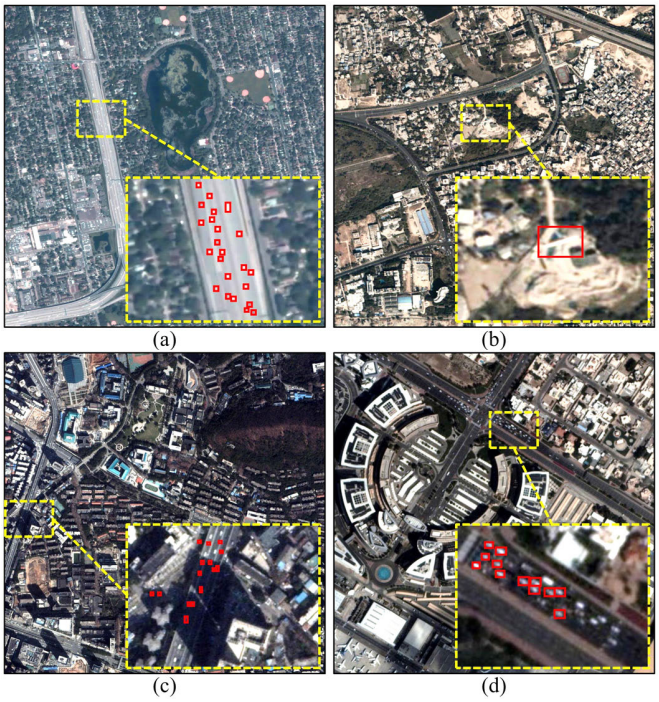


Fig. 1. Visualization of SV scenarios. (a) Weak objects. (b) BC. (c) Partial occlusion and full occlusion. (d) DS.

1) *Data Scarcity:* The scarcity of high-quality SVOT datasets hinders the development of robust representation models, especially for data-driven approaches [4].
2) *Modality Restriction:* Weak objects exhibit increased susceptibility to background interference [5]. Relying on the raw modality may fail in challenging scenarios, particularly under low-light scenes.
3) *Paradigm Gap:* SVOT methods primarily adopt paradigms from correlation filters and Siamese networks [6]. Recent Transformer-based tracking

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advancements have attracted growing attention and set new performance benchmarks.

4) *Underutilization of Multidimensional Features:* The underutilization of spatial and temporal features restricts effective spatiotemporal modeling of objects in SVOT, impeding the capture of critical intraframe/interframe cues [3].

To handle data scarcity, data-driven methods such as TATrans [7], SiamS2F [8], and HRSiam [1] are trained on large-scale generic image/video sets (e.g., ImageNet [9], COCO [10], LaSOT [11], GOT-10k [12], and TrackingNet [13]) before being evaluated on SVOT datasets. However, this may suffer from inferior generalization due to significant differences between GV and SV scenarios [3]. Other methods, like SiamTM [14] and SiamMD [15], perform full fine-tuning on SVOT datasets. However, full fine-tuning is resource-intensive and risks overfitting compared to prompt fine-tuning [16].

While existing methods that track directly on the raw SV modality achieve competitive results, there is still room for improvement. To mitigate modality restriction, methods such as SATDark [5] and MTE-ISVD [17] adjust SV objects by video enhancement before tracking. While these methods show competitive results, they may underperform potentially when the raw modality is discarded, as they rely on the generalization of the enhancement algorithms in diverse scenes.

The paradigm gap remains significant in SVOT. Traditional tracking paradigms, such as correlation filters and Siamese networks, have been widely adopted. For example, several SVOT methods, including HKCF [18], HMTS [19], and CFME [20], are based on the KCF [21]. Others, like DF [22], CPKF [23], DOCPF [24], and MACF [6], are inspired by correlation filters like CSK [25], DSST [26], STRCF [27], Staple [28], and CPF [29], achieving competitive results. State-of-the-art (SOTA) trackers, such as HRLT [30] and RRT [31], built upon the DiMP [32], also demonstrate impressive results. For Siamese-based trackers, methods like REPS [33] and SiamMDM [15] extend the SiamFC [34], while HRSiam [1] and JSANet [35] build upon SiamRPN [36], balancing the accuracy and speed. Recently, Transformer-based trackers have shown remarkable progress in GV tracking [37]. These can be categorized into CNN-Transformer and fully Transformer models [38]. Inspired by CNN-Transformer models (e.g., Stark [39] and TrDiMP [40]), several SVOT algorithms, including SVLPNet [41], TATrans [7], and TransMPP [42], have set new performance ceilings. However, relying on CNNs with local convolution kernels for feature extraction would limit their capability to fully capture global representations.

Multidimensional features can be broadly classified into spatial and temporal features. Spatial features capture appearance representations, using both hand-crafted and deep features [43]. Hand-crafted features, such as HOG [44], color-name [45], Gabor, and color histograms, are widely used to capture texture and structure information of SV objects, as seen in MBLT [46], SATDark [5], and CPKF [23]. Deep appearance features, common in SVOT methods, consist of a shallow layer encoding low-level information with high spatial resolution for precise localization, and a deep layer that captures high-level semantic information invariant to certain object changes [47]. It has shown excellent performance in SVOT methods such as SiamMDM [15], CSLT [48], and SiamS2F [8]. However, CNN may limit the effective representation of deep appearance features within the intraframe context. Temporal features, including both physical and deep temporal components, are essential for capturing interframe dynamics in SVOT, particularly in occlusion scenarios [3]. Physical temporal features, using exponential smoothing, Kalman filter, motion trajectory averaging, and Gaussian mixture models, have been integrated into methods like CFME [20], MACF [6], DOCPF [24], and SATDark [5] to analyze motion patterns. In contrast, deep temporal features, though important, are often underutilized in SVOT methods, such as REPS [33] and SiamS2F [8], which focus more on spatial aspects. To address this gap, several trackers have incorporated dynamic templates (e.g., SiamMDM and TATrans [7]), temporal relation learning (e.g., RRT [31] and HRLT [30]), and temporal propagation (e.g., TransMPP [42]). These methods enable spatial and temporal awareness through end-to-end training but often struggle with manual update strategies and hyperparameters (e.g., updating thresholds and intervals). As a result, there is substantial potential for improvement in spatiotemporal fusion within SVOT.

Inspired by the preceding analysis, we propose STAR, a unified spatiotemporal fusion framework for SVOT. STAR comprises four key components: scene enhancement module (SEM), extraction-correlation-adaptation module (ECAM), temporal decoding module (TDM), and inertial navigation module (INM). SEM adjusts the dynamic range of SV scenes by estimating pixel-wise and high-order curves, generating enhanced multimodal representations. ECAM aligns feature distributions between the pretrained and target domains, facilitating feature extraction, relation learning, and domain adaptation. While TDM incorporates historical object states for current state reasoning end-to-end, refining spatial features through deep temporal feature integration. Finally, the INM leverages physical temporal features to address limitations in deep temporal features during anomalous interferences (e.g., full occlusion (FOC) and DS), ensuring continuous SVOT.

The major contributions of this work are as follows.

STAR, a unified spatiotemporal fusion framework, leverages the prompt learning pattern to mitigate data scarcity with a minimal number of tunable parameters. It not only ensures high efficiency and adaptability in SVOT but also lays a robust foundation for downstream visual tasks.

The SEM and ECAM work in tandem to optimize spatial feature processing. SEM adjusts the dynamic range of SV scenes and generates enhanced multimodal representations. ECAM further refines features. With its local and unified correlation modeling layers, along with the self-modulation refinement adapter (SMRA), ECAM enables comprehensive feature extraction, relation learning, and domain adaptation, effectively capturing spatial information across different SVOT contexts.

For temporal analysis, the TDM integrates historical object states for end-to-end reasoning, incorporating deep temporal

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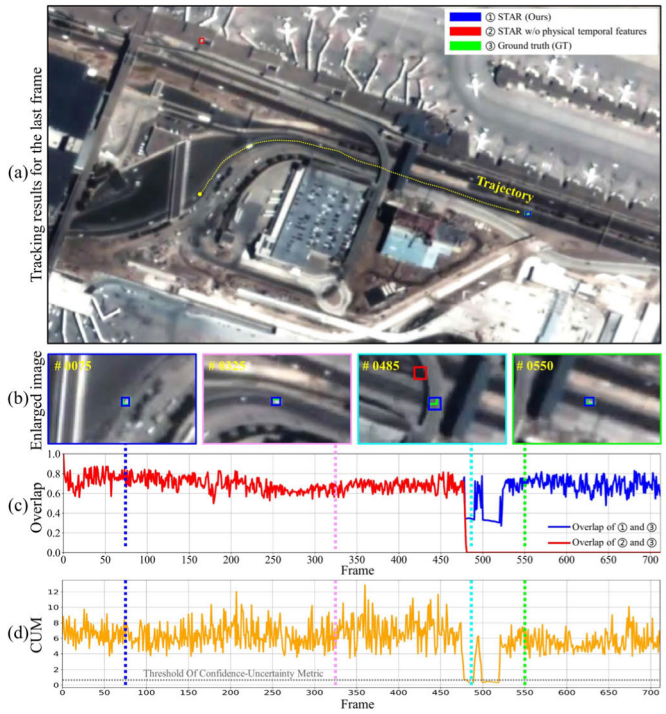


Fig. 2. Spatiotemporal fusion tracking performance analysis. (a) Tracking visualization of the last frame in the 04_000004 (SV248S) sequence. (b) Tracking instances with a zoomed-in view of regional details reveals the limitations of deep temporal features. Discontinuation occurs at frame # 0485 when appearance features become unavailable. (c) Tracking overlaps where full occlusion disrupts temporal continuity. (d) Quantification of the CUM throughout monitoring. STAR demonstrates effective anomaly detection and sustained object pursuit.

features to achieve multidimensional feature representations. By modeling physical temporal features, INM utilizes an awareness selector to assess confidence-uncertainty and an inertial navigation scheme to manage anomalous interferences such as occlusion (see Fig. 2).

Extensive experiments are conducted on popular SVOT benchmarks, namely, SatSOT [4], SV248S [49], and OOTB [3]. Despite varying imaging scenes, resolutions, satellite platforms, and object categories across benchmarks, STAR consistently outperforms nearly 50 SOTA under identical configurations, proving its effectiveness and generalization.

The remainder is structured as follows. Section II reviews related work, Section III presents the method, Section IV details results, and Section V concludes with key findings and future research.

II. Related Work

A. GV Tracking

Visual trackers are generally classified into two categories: generative and discriminative methods [50]. Generative methods build object models and identify matching regions by minimizing an objective loss function. Their effectiveness relies on object representations (e.g., Gaussian mixture model [51] and kernel-based methods [52]) that directly influence both the accuracy and speed. In contrast, discriminative arts focus on distinguishing objects from the background [47]. Core paradigms include correlation filters and Siamese networks, both of which are valued for their simplicity and effectiveness. Correlation filter-based trackers, enhanced with handcrafted features such as HOG [44], color name [45], and Gabor filters, learn discriminative classifiers by minimizing a least-squares loss adaptively. Representative methods in this category include CFLB [53], BACF [54], SAMF [55], SRDCF [56], CSR-DCF [57], AutoTrack [58], and GFS-DCF [59].

However, handcrafted features are often inadequate for representing complex scenarios. Siamese-based methods have further improved performance by using shallow convolutional layers to capture fine spatial details for precise localization, while deeper layers offer more robust semantic information. They utilize a dual-branch structure, one branch for the template (initial frame) and the other for the search image (subsequent frames), with both branches sharing a network for consistent feature extraction.

Reliable object representation is essential for robust tracking, with discriminative models heavily dependent on the backbone network. SiamFC [34], one of the early pioneers of Siamese-based trackers, fine-tunes the AlexNet [60], demonstrating superior results compared to correlation filter-based arts at that time. Several subsequent Siamese-based trackers, including SA-Siam [61], DaSiamRPN [62], SINT [63], and SiamRPN [36], also integrate AlexNet as a feature extractor. However, AlexNet's relatively shallow structure limits its capabilities to extract more powerful features when compared to deeper networks such as ResNeXt [64], VGGNet [65], [66], ResNet [67], and GoogLeNet [68]. Therefore, leveraging wider and deeper networks is critical for improved object discrimination, as seen in SiamRPN++ [69], SiamDW [70], and FEAR [71]. Building on prior advancements, methods have evolved from CNN-Transformer trackers (e.g., TransT [72], Stark [39], TrDiMP [40], and E.T.Track [73]) to two-stream two-stage fully Transformer trackers (e.g., DualTFR [74] and SwinTrack [75]). This progression further leads to the rise of popular one-stream one-stage fully Transformer trackers (e.g., OSTrack [76], ARTrackV2 [77], SBT [78], MixFormer [79], and HiT [80]), marking a milestone in GV tracking.

Indeed, satellite-based tracking benchmarks (e.g., SatSOT [4], SV248S [49], and OOTB [3]) share several challenges, such as small object scales, and complex background interference, with UAV-based tracking benchmarks (e.g., UAV123 [81] and DTB70 [82]). However, SVOT also presents a set of unique difficulties, including ultralow frame rates, significantly larger scene coverage, extremely small objects, and limited temporal continuity. These factors make the direct application of SOTA UAV-based tracking algorithms (e.g., AVTrack [83], SGLATrack [84], and ORTrack [85]) to the satellite domain nontrivial and often ineffective without substantial adaptation.

B. Multimodal Tracking With Prompt Learning

Recent advances in prompt-driven adaptation have improved multimodal tracking capabilities [16], [86]. ProTrack [87] enhances cross-modal integration through foundation model

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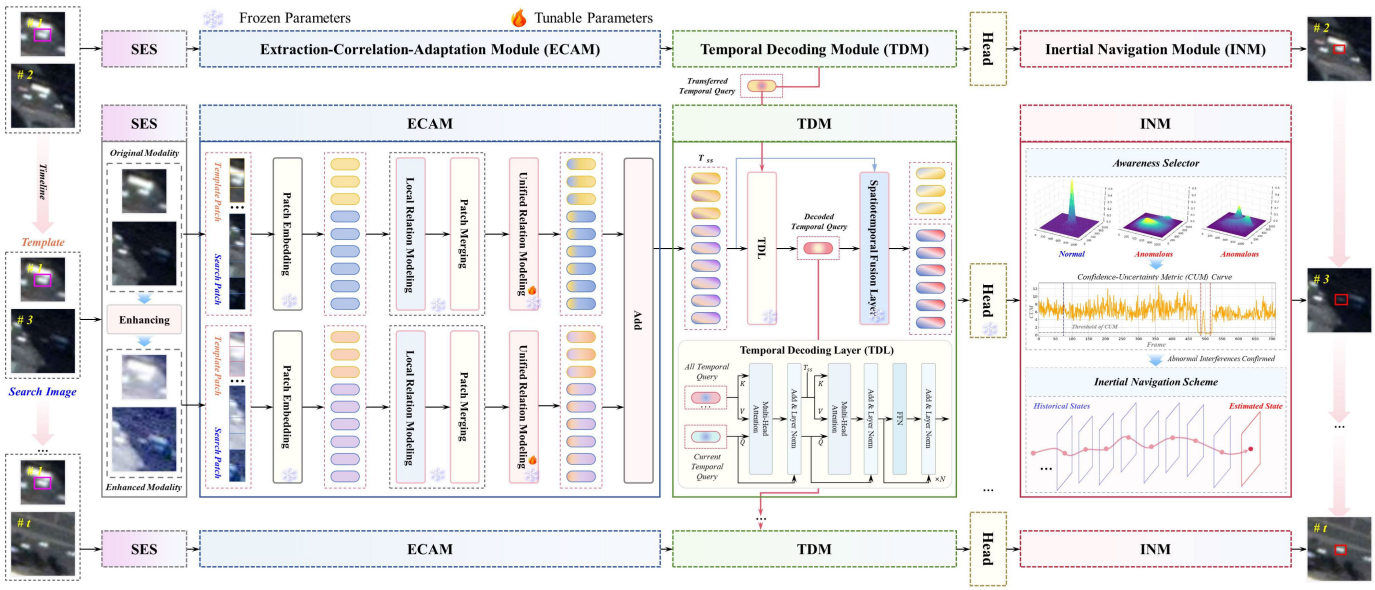


Fig. 3. Overview of the STAR tracker, integrating four principal modules: the SEM for multimodal representation generation, ECAM for facilitating feature extraction, relation learning, and domain adaptation, TDM for enabling end-to-end deep temporal feature embedding, and INM for mitigating anomalous interferences.

alignment. ViPT [88] develops modality-specific prompting mechanisms. BAT [89] employs bidirectional feature alignment to strengthen cross-modal correspondence, whereas TATrack [90] implements temporal-aware modality prompting. SDSTrack [91] demonstrates robustness through limited-data adaptation, and QueryNLT [92] achieves superior discrimination via vision-language co-training. Despite these developments, current research predominantly focuses on GV object tracking (e.g., RGB-E [93], RGB-T [90], RGB-L [92], and RGB-D [94]), with SVOT remaining underexplored. SVOT encounters distinct difficulties (e.g., weak features, FOC, and DS), which are less prevalent in GV tracking. To bridge this gap, we propose a unified spatiotemporal framework for SVOT.

C. Tracking With Spatiotemporal Fusion

Temporal context utilization has become crucial for robust visual tracking [95], [96], [97]. TCTrack++ [98] develops dynamic CNN-Transformer hybrids, while STMTrack [99] utilizes memory-enhanced architectures with regional attention mechanisms. Adaptive template updating mechanisms show effectiveness in managing appearance variations and optimizing cross-frame correspondence, with Stark [39] implementing selective updating for spatiotemporal modeling. SeqTrack [100] and UpdateNet [101] enhance matching accuracy through template optimization strategies. JTBP [102] proposes a propagation framework leveraging temporal consistency for object-background discrimination. Mixformer [79] integrates prediction modules for template refinement, while TATrack [90] adopts dual-stream networks for comprehensive spatiotemporal awareness. MCTrack [103] combines trajectory patterns and frame history for enhanced relation modeling. Current solutions such as ODTrack [104], AQATrack [105], and EVPTrack [106] overcome updating limitations by token-based temporal propagation, eliminating explicit timing updating requirements. These advancements establish theoretical foundations for the proposed STAR, demonstrating that effective spatial–temporal fusion remains vital for SVOT.

III. Methodology

A. Architecture Overview

The STAR consists of four components (see Fig. 3). 1) The SEM optimizes the dynamic ranges of the SV search image ($X \in \mathbb{R}^{H_x \times W_x \times 3}$) and template ($Z \in \mathbb{R}^{H_z \times W_z \times 3}$), generating enhanced representations (search image: $\bar{X} \in \mathbb{R}^{H_x \times W_x \times 3}$, template: $\bar{Z} \in \mathbb{R}^{H_z \times W_z \times 3}$). 2) The ECAM employs hierarchical multimodal Transformers with local and unified relation modeling, jointly achieving feature extraction, relation learning, and domain adaptation. 3) The TDM refines spatial features through deep temporal feature integration, enabling end-to-end historical state reasoning. 4) The INM implements confidence-aware physical temporal modeling through an awareness selector and inertial navigation scheme, mitigating anomalous interferences (e.g., FOC and DS). The spatiotemporal fusion framework enhances robustness in continuous monitoring scenarios.

B. Scene Enhancement Structure (SES)

SVOT presents substantial difficulties, particularly when monitoring small and low-contrast objects in complex scenarios [see Fig. 4(a) and (c)]. These objects frequently demonstrate indistinct characteristics and heightened sensitivity to background interference. The inherent limitations of raw SV modality, typically featured by low-contrast and low-resolution, further compound these challenges by restricting available discriminative features.

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Fig. 4. Comparisons with pre-SES and post-SES processing. Raw modalities are presented in (a) and (c) with post-SES results displayed in (b) and (d). Key rectangular regions of different colors are enlarged and marked (red dashed rectangles) to display enhanced objects and scenes in detail.

Previous studies like OOTB [3] and SATDark [5] have suggested that scene enhancement could effectively address these issues through foreground highlighting and interference suppression while maintaining real-time processing capabilities. STAR extends prompt learning principles by incorporating the optimized enhancer [107] for SV scene enhancements.

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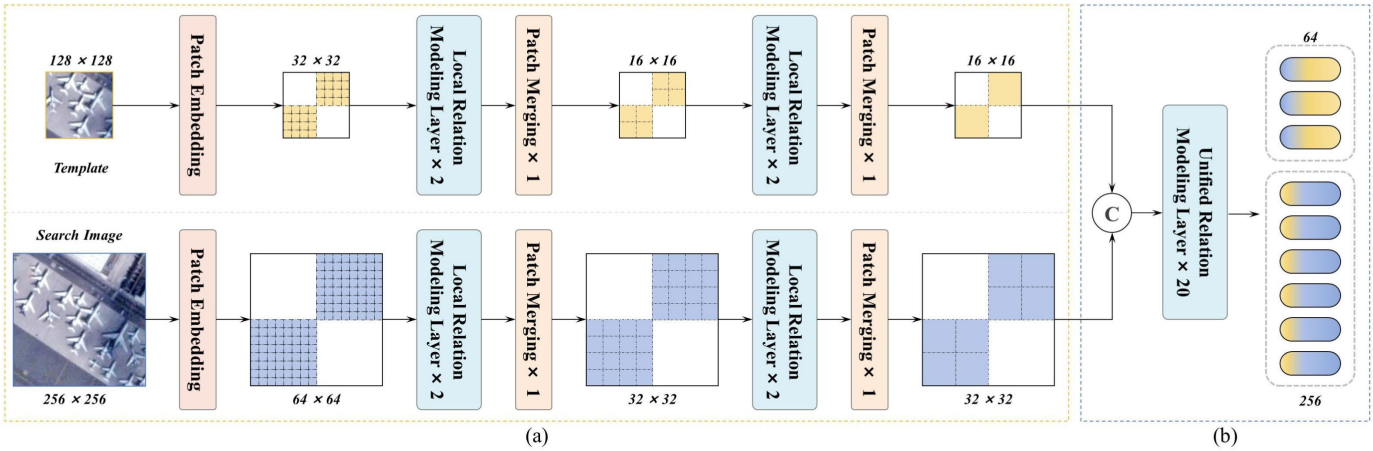


Fig. 5. Structural configuration of the ECAM, integrating two fundamental structures: (a) local relation modeling and (b) unified relation modeling.

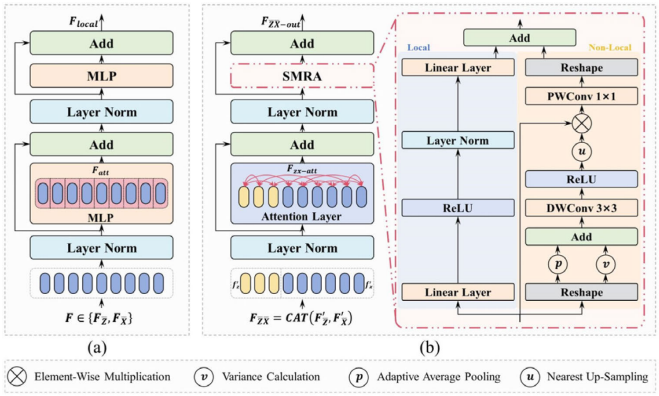


Fig. 6. Illustrations of (a) local relation modeling layer and (b) unified relation modeling layer with SMRA shown in detail.

This lightweight network uses adaptive pixel-wise mapping curves formulated as

$$E(x) = E_{n-1}(x) + \psi(x)E_{n-1}(x)(1 - E_{n-1}(x)) \quad (1)$$

where x is pixel coordinates, $n = 3$ represents iteration cycles, $E(x)$ is the enhanced version of the given position, and ψ is a parameter map.

As demonstrated in Fig. 4, SES significantly improves weak object visibility while preserving structural integrity. The enhanced multi-modal data complements the raw SV modality, forming the input for subsequent modules. This dual-modality integration maintains spatial fidelity while enhancing discriminative features without compromising speed.

C. Extraction-Correlation-Adaptation Module

As depicted in Fig. 5, ECAM comprises a multimodal hierarchical Transformer architecture with independent layer designs for shallow/deep processing stages, integrating local relation modeling and unified relation modeling with the SMRA. This architecture enables simultaneous feature extraction, relation learning, and domain adaptation. Specifically, the initial stage uses stacked local relation modeling layers with patch merging, gradually reducing the token count while expanding channel dimensions [37]. Notably, search image and template processing remain separated in this phase. The subsequent stage implements unified relation modeling for joint feature extraction-correlation-adaptation. ECAM performs joint processing of the enhanced modality (from SES) and raw SV modality through shared-weight layers with SMRA embedded.

TABLE I
DETAILS OF THE TRACKING MODEL

Tracker	Backbone	Resolution	TMPs	Params	FLOPs	FPS
STAR	HiViT-Base	256×256	0.794M	72.784M	50.974G	87.3

Below, we detail components of the local relation modeling and unified relation modeling.

1) Local Relation Modeling: Fig. 6(a) illustrates the structure of the shallow-stage local relation modeling, replacing standard multihead attention with MLP-based local feature enrichment

$$\begin{cases} F_{\text{att}} = F \oplus \text{MLP}(\text{LayerNorm}(f)), & F \in \{F_{\bar{Z}}, F_{\bar{X}}\} \\ F_{\text{local}} = F_{\text{att}} \oplus \text{MLP}(\text{LayerNorm}(F_{\text{att}})) \end{cases} \quad (2)$$

where $F_{\bar{Z}}$ and $F_{\bar{X}}$ denote template and search image tokens (using the enhanced modality as an example) processed through patch embedding, positional encoding, and reshaping operations. F_{local} is the output of the local relation modeling layer. Channel-wise convolution supplements shallow feature expression, with cross-image modeling blocked for efficiency. To optimize efficiency while maintaining feature hierarchy, two local relation modeling layers with subsequent patch merging operations are configured in the shallow stages.

2) Unified Relation Modeling: The unified modeling layer [see Fig. 6(b)] unifies intraimage and interimage relation modeling within a single architecture. This architecture builds upon standard Transformer designs [108], integrating vanilla global attention mechanisms [109] with SMRA for feature distribution adaptation (see Fig. 6). Especially, the unified relation modeling jointly processes the template ($F'_{\bar{Z}}$) and

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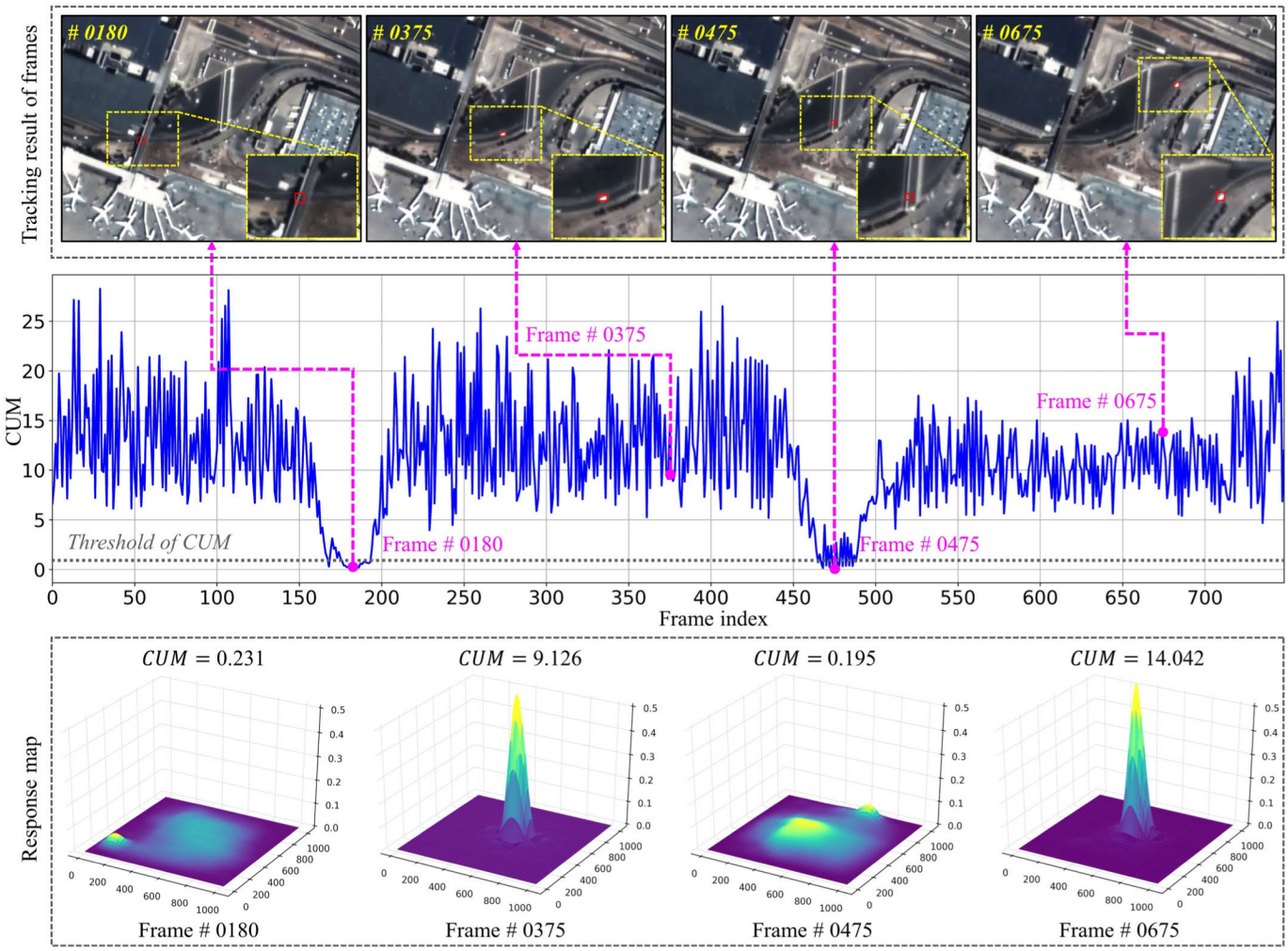


Fig. 7. CUM analysis illustrated by the car_01 (SatSOT) sequence. Notable CUM decreases indicate the activation of the inertial navigation scheme. Tracking outcomes are presented in the top, with magnified views of yellow-bounded regions. The middle illustrates the CUM, featuring a threshold τ marked by a gray dashed line. Corresponding response maps are visualized at the bottom. Observations reveal spatial feature instability at frames # 0180 and # 0475, attributable to the full occlusion. Under such a scenario, response maps exhibit flattened, multipeak distributions, yielding reduced CUM values. Consequently, anomalous interferences can be identified through CUM evaluation.

search image ($F'_{\bar{X}}$) after local relation modeling

$$F_{\bar{Z}\bar{X}} = \mathrm{CAT}\left(F'_{\bar{Z}}, F'_{\bar{X}}\right) \tag{3}$$

$$\begin{cases} F_{\bar{Z}\bar{X}\text{-att}} = F_{\bar{Z}\bar{X}} \oplus \mathrm{MHSA}(\mathrm{LayerNorm}(F_{\bar{Z}\bar{X}})) \\ F_{\bar{Z}\bar{X}\text{-out}} = F_{\bar{Z}\bar{X}\text{-att}} \oplus \mathrm{SMRA}\left(\mathrm{LayerNorm}\left(F_{\bar{Z}\bar{X}\text{-att}}\right)\right) \end{cases} \tag{4}$$

where $F_{\bar{Z}\bar{X}}$ and $F_{\bar{Z}\bar{X}\text{-out}}$ are the input and the output of the unified relation modeling layer, respectively. The SMRA (see Fig. 6) processes the input token ($F_{\mathrm{in}} \in \mathbb{R}^{N\times C}$) using a standard MLP structure for estimating local details while a self-modulation structure for nonlocal information exploration. The MLP branch employs sequential linear transformations with ReLU activation (δ) to capture localized patterns

$$F_{\mathrm{mlp}} = \mathrm{Linear}(\delta(\mathrm{LayerNorm}(\mathrm{Linear}(F_{\mathrm{in}})))) \tag{5}$$

where F_{mlp} denotes the captured local details.

Simultaneously, the self-modulation branch integrates spatial statistics by the depth-wise convolution. We compute local window variance (σ^2) and adaptive average pooling (p) of reshaped features ($R(F_{\mathrm{in}})$), subsequently combining them via

$$F_{\mathrm{sm}} = u\left(\delta\left(\mathrm{DWConv}\left(\sigma^2(R(F_{\mathrm{in}})) \oplus p(R(F_{\mathrm{in}}))\right)\right)\right) \tag{6}$$

where R denotes the reshape operation and DWConv refers to the depth-wise 1-D convolution. F_{sm} denotes the modulated feature, facilitating effective information aggregation by

$$F_{m\text{-out}} = R(\mathrm{PWConv}(F_{\mathrm{in}} \otimes F_{\mathrm{sm}})) \tag{7}$$

where PWConv corresponds to the pixel-wise convolution.

Finally, the hierarchical feature fusion mechanism combines local precision and global context through ($F_{\mathrm{mlp}} \oplus F_{m\text{-out}}$), establishing a unified feature relation framework.

D. Temporal Decoding Module

Spatial analysis alone proves insufficient for addressing persistent tracking challenges such as dynamic motion patterns. Therefore, STAR incorporates a TDM (see Fig. 3) that integrates deep temporal features via attention propagation. TDM contains a temporal decoding layer for distilling time-series

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TABLE II
PERFORMANCE ASSESSMENT OF GV TRACKERS ON SATSOT AND SV248S

Trackers	Venues	Features/Backbones	Tags	SatSOT		SV248S	
				PR	SR	PR	SR
CSK [25]	*ECCV 2012*	I	Default	0.198	0.229	0.115	0.077
CN [112]	*CVPR 2014*	CN+I	Default	0.350	0.335	0.142	0.086
DAT [113]	*CVPR 2015*	CH	Default	0.421	0.350	0.390	0.238
SRDCF [56]	*ICCV 2015*	HOG	Default	0.561	0.425	0.686	0.331
Staple [28]	*CVPR 2016*	HOG+CN	Default	0.530	0.416	**0.750**	**0.481**
BACF [54]	*ICCV 2017*	HOG	Default	0.524	0.383	0.446	0.151
CSRDCF [57]	*IJCV 2018*	HOG+CN+CH	Default	0.546	0.440	0.693	0.407
STRCF [27]	*CVPR 2018*	HOG+CN	Default	0.548	0.414	**0.704**	0.354
SiamRPN [36]	*CVPR 2018*	AlexNet	Default	0.377	0.316	0.330	0.145
DaSiamRPN [62]	*ECCV 2018*	AlexNet	VOT	0.460	0.366	0.520	0.203
SiamRPN++ [69]	*CVPR 2019*	ResNet-50	Default	0.543	0.411	0.563	0.231
UpdateNet [101]	*ICCV 2019*	AlexNet	DaSiamRPN	0.509	0.407	0.560	0.218
SiamDW [70]	*CVPR 2019*	CIRIncep22	SiamFC	0.507	0.408	0.611	0.393
SiamMask [114]	*CVPR 2019*	ResNet-50	HBB	0.548	0.398	0.592	0.232
SiamBAN [115]	*CVPR 2020*	ResNet-50	OTB	0.503	0.356	0.465	0.184
SiamFC++ [116]	*AAAI 2020*	GoogLeNet	OTB	0.489	0.376	0.431	0.173
SiamGAT [117]	*CVPR 2021*	GoogLeNet	GOT-10k	0.510	0.393	0.620	0.381
LightTrack [118]	*CVPR 2021*	Custom	Mobile	0.373	0.309	0.201	0.112
TransT [72]	*CVPR 2021*	ResNet-50	Default	0.487	0.379	0.381	0.173
Stark [39]	*ICCV 2021*	ResNet-50	ST50	0.387	0.328	0.257	0.130
SiamCAR [119]	*IJCV 2022*	ResNet-50	Default	**0.587**	0.441	0.697	**0.433**
OSTrack [76]	*ECCV 2022*	ViT-Base	256-Default	0.475	0.390	0.314	0.152
SimTrack [120]	*ECCV 2022*	ViT-Base	ViT-B/16	0.327	0.304	0.167	0.104
SBT [78]	*CVPR 2022*	SBT-Base	Default	0.469	0.385	0.385	0.172
TCTrack [121]	*CVPR 2022*	AlexNet	Default	0.428	0.322	0.311	0.135
GRM [122]	*CVPR 2023*	ViT-Base	256-Default	0.444	0.369	0.288	0.141
SeqTrack [100]	*CVPR 2023*	ViT-Base	256-Default	0.490	0.408	0.366	0.161
TCTrack++ [98]	*TPAMI 2023*	AlexNet	Default	0.492	0.362	0.381	0.165
ARTrack [123]	*CVPR 2023*	ViT-Base	256-Default	0.564	**0.443**	0.463	0.199
HiT [80]	*ICCV 2023*	LeViT-Base	384-Default	0.287	0.255	0.130	0.080
AVTrack [83]	*ICML 2024*	ViT-Tiny	224-Default	0.444	0.381	0.364	0.177
ODTrack [104]	*AAAI 2024*	ViT-Base	384-Default	0.524	0.419	0.438	0.182
SMAT [124]	*WACV 2024*	MobileViTv2	Default	0.414	0.337	0.254	0.139
AQATrack [105]	*CVPR 2024*	HiViT-Base	256-Default	0.533	0.417	0.465	0.200
EVPTrack [106]	*AAAI 2024*	HiViT-Base	224-Default	0.501	0.406	0.461	0.194
HIPTrack [125]	*CVPR 2024*	ViT-Base	Default	0.526	0.422	0.331	0.153
ARTrackV2 [77]	*CVPR 2024*	ViT-Base	256-Default	**0.626**	**0.478**	0.524	0.224
SGLATrack [84]	*CVPR 2025*	DeiT-tiny	256-Default	0.483	0.391	0.409	0.186
ORTrack [85]	*CVPR 2025*	DeiT-tiny	256-Default	0.460	0.380	0.382	0.175
STAR	*Ours*	HiViT-Base	Default	**0.639**(+1.3%)	**0.537**(+5.9%)	**0.774**(+2.4%)	**0.523**(+4.2%)

Top-performing metrics are color-coded: red (1st), green (2nd), and blue (3rd). STAR's improvements over the second-best method in each column are highlighted.

patterns from sequential data and a spatiotemporal fusion layer for combining spatial–temporal characteristics. TDM monitors object state evolution and motion pattern transitions, with processed tokens subsequently directed to the prediction network for localization.

E. Inertial Navigation Module

While TDM can handle partial occlusion (POC) scenarios, its efficacy diminishes during FOC. To this end, we devise the INM that leverages both the visual and motion information for enhanced robustness. It models physical temporal

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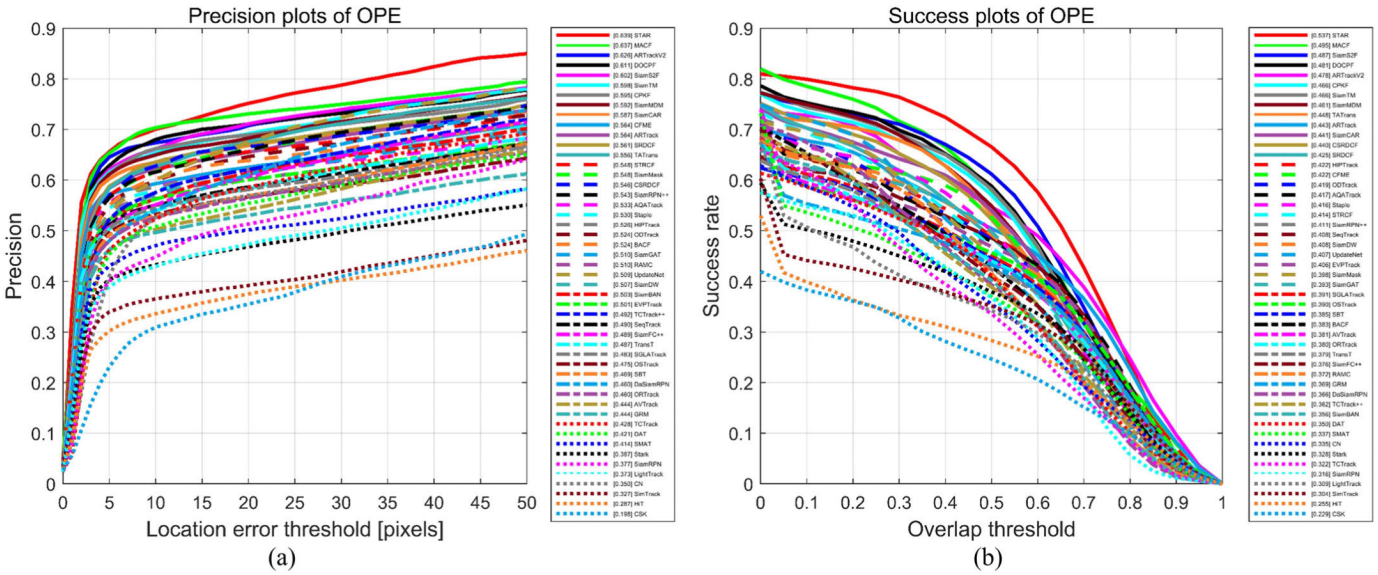


Fig. 8. Performance comparisons with GV and SV trackers on SatSOT. (a) Precision plot. (b) Success plot.

features, including an awareness selector to assess the tracking confidence-uncertainty and an inertial navigation scheme to manage anomalous interferences. For each frame, visual and deep temporal features are first utilized to locate the object, followed by physical temporal feature integration to refine the position when visual cues are insufficient.

1) Awareness Selector: We present the awareness selector that dynamically evaluates the tracking confidence-uncertainty for sensing anomalous interferences. The confidence-uncertainty can be inferred from response map characteristics. A precise match typically yields a distinct peak with minimal surrounding fluctuations, whereas inaccurate detections produce erratic patterns (see Fig. 7) [22], [23]. Building upon this observation, the confidence-uncertainty metric (CUM) is introduced based on peak intensity and peak-to-sidelobe ratio [6]. The awareness selector computes the CUM derived from response maps

$$\mathrm{CUM}(\mathrm{RM}) = \frac{(\mathrm{Max}(\mathrm{RM}) - \mathrm{Avg}(\mathrm{RM})) \times \mathrm{Max}(\mathrm{RM})}{\mathrm{Std}(\mathrm{RM})} \tag{8}$$

where Max, Avg, and Std represent the maximum, average, and standard deviation of the response map RM, respectively.

For the current frame t, let $\mathrm{CUM}_t^{\mathrm{avg}}(\mathrm{RM})$ denote the average CUM across k recent reliable frames. A CUM threshold τ is used to determine the activation condition for the inertial navigation scheme by meeting

$$\mathrm{CUM}_t(\mathrm{RM}) \big/ \mathrm{CUM}_t^{\mathrm{avg}}(\mathrm{RM}) < \tau. \tag{9}$$

Fig. 7 illustrates CUM variations during SVOT, demonstrating its role in identifying interferences and maintaining accurate object trajectories.

2) Inertial Navigation Scheme: The inertial navigation scheme can be activated upon meeting (9). In SV, objects typically exhibit smooth motion trajectories due to the bird's-eye perspective [6], [24]. The higher order models or adaptive filters (e.g., particle filters) can theoretically capture more complex motions. However, they may introduce additional

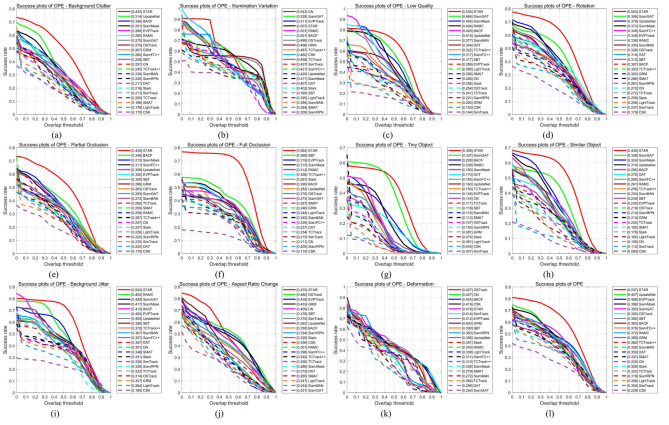


Fig. 9. Attribute-wise success plot on SatSOT. (a)–(k) Individual attributes. (l) OVE evaluation.

computational overhead, which may restrict real-time applications in SVs. Therefore, nonlinear regression is used to model motion patterns and predict object positions [22]. Let $(t_i, \varphi_i), i = 1, 2, 3, \ldots, p$ represent the object's x-axis coordinates across p frames. The smooth trajectory is modeled using a quadratic function

$$\varphi_i = \beta_0 + \beta_1 t_i + \beta_2 t_i^2 \tag{10}$$

where β_0, β_1, and β_2 are determined by

$$\min \sum_{i=1}^{p} \left(\beta_0 + \beta_1 t_i + \beta_2 t_i^2 - \varphi_i\right)^2. \tag{11}$$

Solving for $[\beta_0, \beta_1, \beta_2]^{\mathrm{T}}$ provides the trajectory equation. The y-axis coordinates are computed similarly, yielding the final position. Notably, the CUM remains unchanged when using an inertial navigation scheme, as actual measurements are unavailable. Conversely, the CUM can be updated, thereby improving SVOT accuracy.

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TABLE III
PERFORMANCE ASSESSMENT OF SV TRACKERS ON SATSOT AND SV248S

No.	Trackers	Venues	Features/Backbones	POP	FPS	SatSOT		SV248S	
						PR	SR	PR	SR
1	CFME [20]	*TGRS 2020*	HOG	CPU	**123.0**	0.564	0.422	0.426	0.290
2	RAMC [126]	*RS 2022*	HOG+OF	CPU	42.4	0.510	0.372	0.707	0.384
3	CPKF [23]	*TGRS 2022*	HOG+CN+GCS	CPU	25.3	0.595	0.466	-	-
4	SiamMDM [15]	*TGRS 2023*	AlexNet	GPU	**110.4**	0.592	0.461	0.711	0.453
5	SiamTM [14]	*RS 2023*	ResNet-50	GPU	**89.7**	0.598	0.466	**0.746**	**0.487**
6	SVLPNet [41]	*TCSVT 2024*	ResNet-50	GPU	-	-	-	0.688	0.471
7	DOCPF [24]	*TGRS 2024*	HOG+CN+GCS	CPU	34.6	**0.611**	0.481	-	-
8	MACF [6]	*TGRS 2024*	HOG+CN+CH	GPU	12.0	**0.637**	**0.495**	-	-
9	TATrans [7]	*TGRS 2024*	ResNet-18	GPU	26.4	0.556	0.448	0.618	0.339
10	SiamS2F [8]	*TGRS 2025*	ResNet-50	GPU	82.2	0.602	**0.487**	**0.757**	**0.506**
11	STAR	*Ours*	HiViT-Base	GPU	87.3	**0.639**(+0.2%)	**0.537**(+4.2%)	**0.774**(+1.7%)	**0.523**(+1.7%)

POP is the primary operational platform. FPS indicates the authorized tracking speed on the SatSOT evaluation. STAR's improvements over the second-best method in each column are highlighted.

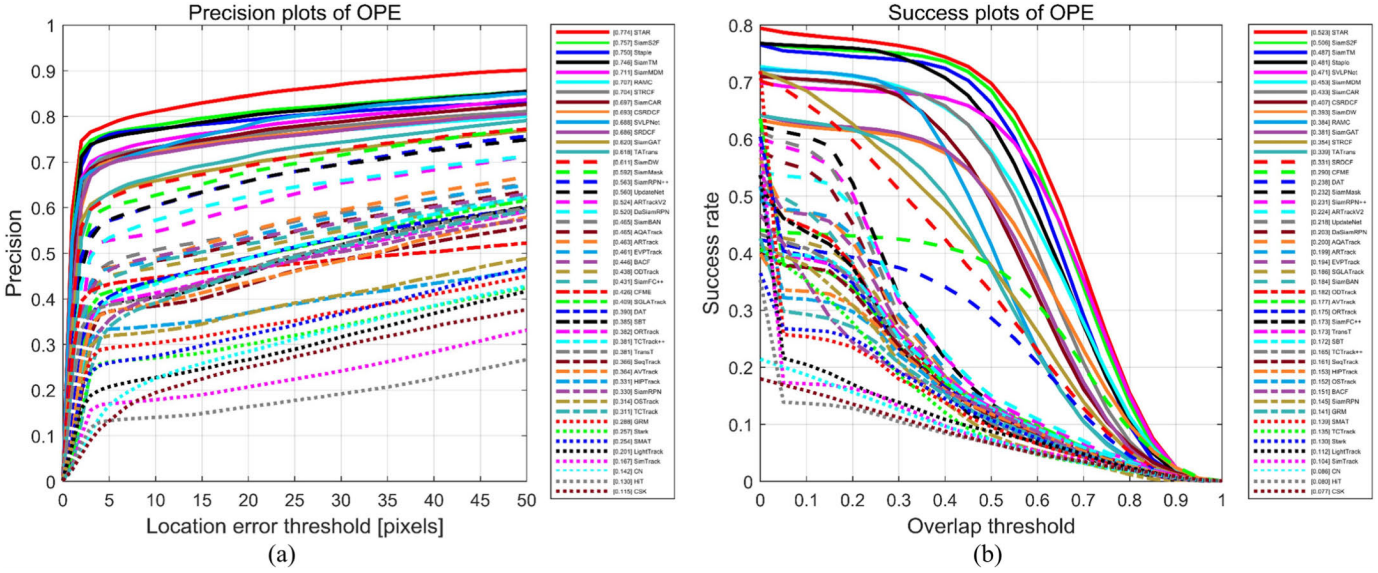


Fig. 10. Performance comparisons with GV and SV trackers on SV248S. (a) Precision plot. (b) Success plot.

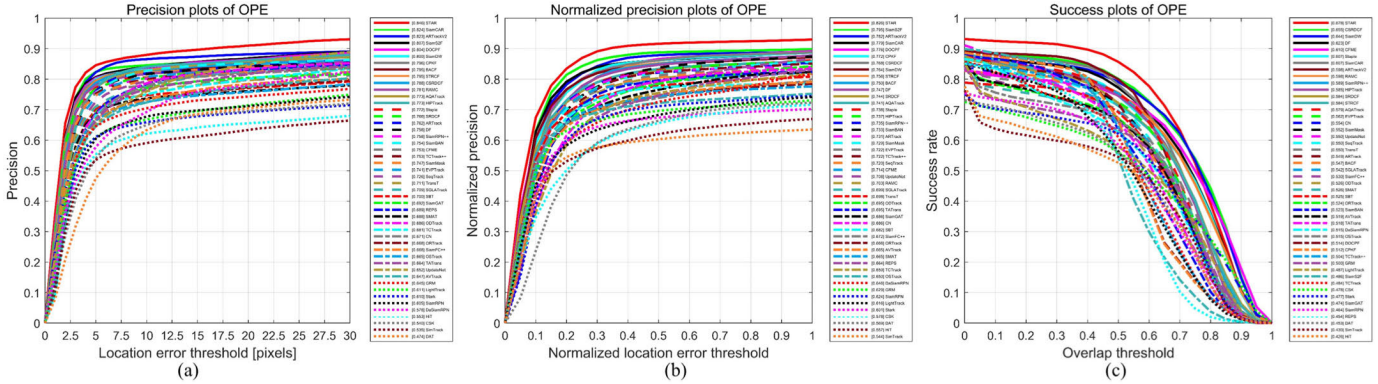


Fig. 11. Performance comparisons with GV and SV trackers on OOTB. (a) Precision plot. (b) Normalized precision plot. (c) Success plot.

F. Head and Loss

STAR processes search region tokens by transforming them into a 2-D spatial representation for the head network, which comprises a classification branch for initial object localization, a local offset branch for precise center adjustment, and a size estimation branch for bounding box size prediction [76].

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TABLE IV
CATEGORY-WISE AND DIFFICULTY-WISE PR AND SR EVALUATIONS ON SV248S

Trackers	Venues	Category-Wise Evaluations								Difficulty-Wise Evaluations					
		Vehicle		L-Vehicle		Airplane		Ship		Simple		Normal		Hard	
		PR	SR	PR	SR	PR	SR	PR	SR	PR	SR	PR	SR	PR	SR
CSK [25]	*ECCV 2012*	0.093	0.047	0.140	0.128	0.608	0.649	0.332	0.278	0.140	0.107	0.100	0.057	0.080	0.040
CN [112]	*CVPR 2014*	0.120	0.058	0.179	0.136	0.629	0.686	0.207	0.177	0.171	0.116	0.120	0.066	0.112	0.050
DAT [113]	*CVPR 2015*	0.322	0.183	0.723	0.463	0.645	0.685	0.359	0.284	0.534	0.336	0.273	0.160	0.285	0.160
SRDCF [56]	*ICCV 2015*	0.648	0.281	0.850	0.521	0.798	0.719	0.992	0.532	0.849	0.446	0.569	0.242	0.511	0.226
Staple [28]	*CVPR 2016*	**0.731**	0.456	0.835	0.568	0.761	0.696	0.954	0.588	**0.956**	**0.624**	0.585	0.365	**0.589**	0.368
BACF [54]	*ICCV 2017*	0.409	0.104	0.577	0.288	0.757	0.721	0.728	0.447	0.563	0.211	0.346	0.107	0.378	0.087
CSRDCF [57]	*IJCV 2018*	0.667	0.376	0.798	0.505	0.788	0.751	0.995	0.632	0.894	0.535	0.521	0.300	0.575	0.324
STRCF [27]	*CVPR 2018*	0.672	0.317	0.845	0.486	0.781	0.672	0.993	0.586	0.894	0.474	0.549	0.261	0.565	0.248
SiamRPN [36]	*CVPR 2018*	0.283	0.106	0.477	0.248	0.719	0.630	0.913	0.576	0.486	0.214	0.220	0.098	0.159	0.065
DaSiamRPN [62]	*ECCV 2018*	0.487	0.156	0.636	0.336	0.673	0.733	**1.000**	0.631	0.667	0.270	0.422	0.160	0.334	0.112
SiamRPN++ [69]	*CVPR 2019*	0.507	0.167	0.763	0.448	**0.989**	**0.835**	0.997	0.636	0.744	0.320	0.458	0.177	0.284	0.104
UpdateNet [101]	*ICCV 2019*	0.510	0.165	0.763	0.398	0.759	0.734	**1.000**	0.573	0.740	0.299	0.447	0.169	0.307	0.101
SiamDW [70]	*CVPR 2019*	0.568	0.354	0.778	0.530	0.858	0.807	0.953	0.566	0.778	0.508	0.491	0.308	0.431	0.283
SiamMask [114]	*CVPR 2019*	0.542	0.173	0.773	0.441	0.935	0.727	0.991	0.583	0.756	0.308	0.498	0.185	0.330	0.118
SiamBAN [115]	*CVPR 2020*	0.398	0.124	0.730	0.391	0.974	0.815	0.695	0.420	0.623	0.256	0.367	0.137	0.245	0.091
SiamFC++ [116]	*AAAI 2020*	0.386	0.128	0.573	0.298	0.974	0.798	0.641	0.415	0.573	0.238	0.344	0.134	0.226	0.080
SiamGAT [117]	*CVPR 2021*	0.615	0.366	0.569	0.380	0.929	0.768	0.983	0.618	0.751	0.474	0.516	0.308	0.515	0.301
LightTrack [118]	*CVPR 2021*	0.126	0.060	0.446	0.260	0.954	0.785	0.694	0.497	0.276	0.157	0.148	0.084	0.111	0.053
TransT [72]	*CVPR 2021*	0.309	0.112	0.637	0.368	0.928	**0.829**	0.976	0.588	0.517	0.243	0.289	0.130	0.212	0.075
Stark [39]	*ICCV 2021*	0.179	0.070	0.549	0.324	0.960	0.788	0.541	0.469	0.371	0.194	0.162	0.078	0.182	0.082
SiamCAR [119]	*IJCV 2022*	0.663	0.401	0.812	0.543	**0.985**	0.756	**0.998**	0.587	0.866	0.548	0.573	0.348	0.524	0.317
OSTrack [76]	*ECCV 2022*	0.245	0.097	0.560	0.321	0.964	0.809	0.648	0.432	0.449	0.216	0.207	0.102	0.209	0.097
SimTrack [120]	*ECCV 2022*	0.111	0.064	0.318	0.194	0.923	0.765	0.576	0.384	0.220	0.139	0.116	0.073	0.159	0.089
SBT [78]	*CVPR 2022*	0.307	0.104	0.672	0.407	0.951	0.801	0.993	0.596	0.515	0.244	0.280	0.118	0.286	0.106
TCTrack [121]	*CVPR 2022*	0.244	0.081	0.522	0.290	0.949	0.801	0.962	0.521	0.403	0.185	0.257	0.106	0.172	0.056
GRM [122]	*CVPR 2023*	0.208	0.081	0.586	0.336	0.962	0.791	0.665	0.449	0.402	0.202	0.192	0.092	0.214	0.091
SeqTrack [100]	*CVPR 2023*	0.290	0.099	0.636	0.367	**0.983**	0.804	0.914	0.544	0.474	0.223	0.282	0.115	0.268	0.099
TCTrack++ [98]	*TPAMI 2023*	0.317	0.108	0.588	0.342	0.951	0.810	**0.999**	0.556	0.514	0.231	0.290	0.120	0.225	0.088
ARTrack [123]	*CVPR 2023*	0.398	0.135	0.693	0.415	0.944	0.806	**0.998**	0.601	0.602	0.273	0.362	0.146	0.315	0.120
HiT [80]	*ICCV 2023*	0.072	0.038	0.294	0.176	0.875	0.766	0.529	0.348	0.182	0.114	0.088	0.054	0.092	0.049
AVTrack [83]	*ICML 2024*	0.291	0.114	0.656	0.405	0.895	0.746	0.667	0.479	0.508	0.247	0.257	0.128	0.223	0.100
ODTrack [104]	*AAAI 2024*	0.363	0.117	0.747	0.418	0.932	0.796	0.673	0.413	0.584	0.255	0.341	0.132	0.252	0.094
SMAT [124]	*WACV 2024*	0.184	0.089	0.466	0.276	0.947	0.797	0.954	0.556	0.357	0.196	0.183	0.101	0.135	0.072
AQATrack [105]	*CVPR 2024*	0.402	0.138	0.690	0.408	0.939	0.778	0.971	0.597	0.607	0.272	0.357	0.147	0.333	0.125
EVPTrack [106]	*AAAI 2024*	0.408	0.137	0.656	0.389	0.963	0.803	0.681	0.455	0.613	0.267	0.332	0.136	0.369	0.135
HIPTrack [125]	*CVPR 2024*	0.253	0.091	0.626	0.364	0.967	0.807	0.682	0.480	0.447	0.220	0.236	0.102	0.247	0.097
ARTrackV2 [77]	*CVPR 2024*	0.459	0.157	0.772	0.460	0.958	0.822	**1.000**	0.616	0.704	0.308	0.399	0.167	0.316	0.118
SGLATrack [84]	*CVPR 2025*	0.338	0.123	0.683	0.407	0.961	0.800	0.699	0.494	0.567	0.262	0.298	0.134	0.233	0.097
ORTrack [85]	*CVPR 2025*	0.322	0.120	0.608	0.360	0.906	0.776	0.574	0.405	0.546	0.248	0.266	0.125	0.203	0.090
CFME [20]	*TGRS 2020*	0.381	0.247	0.592	0.428	0.788	0.765	0.697	0.486	0.496	0.347	0.409	0.267	0.239	0.165
RAMC [126]	*RS 2022*	0.675	0.344	0.846	0.533	0.781	0.739	**0.998**	0.580	0.893	0.497	0.575	0.296	0.505	0.287
SiamMDM [15]	*TGRS 2023*	0.679	0.420	0.833	0.581	0.897	0.719	**1.000**	0.592	0.893	0.576	0.581	0.362	0.514	0.334
SiamTM [14]	*RS 2023*	0.723	**0.466**	**0.867**	0.572	0.950	0.749	0.382	0.277	**0.900**	0.602	**0.626**	**0.398**	**0.617**	**0.384**
SVLPNet [41]	*TCSVT 2024*	0.650	0.432	0.827	**0.609**	0.958	0.823	0.977	**0.691**	0.842	0.593	0.588	0.390	0.487	0.320
TATrans [7]	*TGRS 2024*	0.571	0.295	0.818	0.488	0.784	0.706	**0.998**	0.661	0.781	0.442	0.491	0.259	0.481	0.247
SiamS2F [8]	*TGRS 2025*	**0.729**	**0.473**	**0.854**	**0.615**	0.967	**0.825**	**1.000**	**0.713**	**0.926**	**0.633**	**0.632**	**0.413**	0.587	**0.375**
STAR	*Ours*	**0.751**	**0.490**	**0.863**	**0.645**	0.864	0.765	**0.999**	**0.744**	0.879	**0.617**	**0.705**	**0.460**	**0.640**	**0.406**

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TABLE V
ATTRIBUTE-WISE AND OVE PR/SR EVALUATIONS ON SV248S

Trackers	STO	LTO	DS	IV	BCH	SM	ND	CO	BCL	IPR	OVE
CSK [25]	0.107/0.060	0.115/0.077	0.099/0.060	0.102/0.061	0.103/0.076	0.129/0.078	0.071/0.035	0.123/0.068	0.088/0.044	0.110/0.067	0.115/0.077
CN [112]	0.130/0.070	0.153/0.106	0.128/0.071	0.122/0.066	0.135/0.087	0.165/0.091	0.126/0.062	0.129/0.066	0.103/0.047	0.134/0.076	0.142/0.086
DAT [113]	0.273/0.162	0.222/0.139	0.397/0.234	0.247/0.150	0.392/0.245	0.360/0.208	0.400/0.237	0.242/0.142	0.272/0.152	0.330/0.187	0.390/0.238
BACF [54]	0.332/0.106	0.312/0.137	0.432/0.132	0.402/0.121	0.426/0.163	0.522/0.138	0.439/0.118	0.259/0.092	0.340/0.086	0.399/0.120	0.446/0.151
SiamRPN [36]	0.221/0.097	0.180/0.109	0.336/0.135	0.312/0.126	0.305/0.138	0.278/0.128	0.224/0.099	0.213/0.091	0.184/0.073	0.217/0.100	0.330/0.145
UpdateNet [101]	0.404/0.156	0.242/0.123	0.553/0.199	0.472/0.173	0.552/0.232	0.513/0.187	0.421/0.122	0.340/0.126	0.414/0.135	0.467/0.172	0.560/0.218
SiamMask [114]	0.426/0.161	0.332/0.157	0.572/0.212	0.476/0.168	0.563/0.248	0.497/0.191	0.409/0.153	0.329/0.122	0.476/0.162	0.512/0.187	0.592/0.232
SiamBAN [115]	0.364/0.142	0.266/0.138	0.459/0.168	0.337/0.119	0.473/0.208	0.354/0.133	0.334/0.131	0.266/0.099	0.294/0.096	0.410/0.147	0.465/0.184
SiamFC++ [116]	0.325/0.128	0.372/0.170	0.413/0.155	0.356/0.128	0.411/0.181	0.291/0.114	0.204/0.073	0.287/0.113	0.281/0.097	0.361/0.136	0.431/0.173
SiamGAT [117]	**0.449**/**0.273**	**0.494**/**0.315**	**0.610**/**0.368**	**0.562**/**0.340**	**0.580**/**0.355**	**0.562**/**0.335**	**0.565**/**0.325**	**0.368**/**0.227**	**0.555**/**0.319**	**0.587**/**0.352**	**0.620**/**0.381**
LightTrack [118]	0.158/0.087	0.179/0.108	0.178/0.092	0.137/0.067	0.267/0.152	0.102/0.065	0.168/0.092	0.133/0.073	0.105/0.052	0.141/0.072	0.201/0.112
Stark [39]	0.170/0.079	0.224/0.124	0.232/0.110	0.185/0.089	0.291/0.156	0.206/0.105	0.175/0.100	0.138/0.070	0.151/0.068	0.193/0.091	0.257/0.130
OSTrack [76]	0.208/0.100	0.264/0.148	0.287/0.132	0.243/0.114	0.350/0.176	0.200/0.107	0.225/0.123	0.186/0.087	0.189/0.089	0.257/0.114	0.314/0.152
SimTrack [120]	0.119/0.073	0.180/0.126	0.132/0.082	0.105/0.074	0.193/0.117	0.151/0.098	0.129/0.086	0.123/0.073	0.122/0.073	0.154/0.085	0.167/0.104
SBT [78]	0.267/0.117	0.328/0.166	0.348/0.148	0.299/0.122	0.396/0.194	0.317/0.135	0.274/0.123	0.213/0.093	0.268/0.097	0.298/0.118	0.385/0.172
TCTrack [121]	0.253/0.103	0.252/0.133	0.284/0.114	0.225/0.095	0.363/0.168	0.259/0.108	0.133/0.058	0.206/0.084	0.197/0.069	0.250/0.097	0.311/0.135
GRM [122]	0.200/0.094	0.280/0.149	0.262/0.119	0.206/0.094	0.336/0.169	0.201/0.099	0.210/0.114	0.170/0.081	0.179/0.077	0.218/0.100	0.288/0.141
TCTrack++ [98]	0.283/0.121	0.264/0.142	0.352/0.141	0.280/0.118	0.418/0.191	0.307/0.129	0.201/0.095	0.214/0.087	0.242/0.088	0.321/0.123	0.381/0.165
SMAT [124]	0.185/0.096	0.191/0.138	0.226/0.119	0.224/0.112	0.279/0.156	0.198/0.112	0.140/0.080	0.190/0.099	0.135/0.076	0.203/0.109	0.254/0.139
EVPTrack [106]	0.313/0.132	0.374/0.174	0.431/0.173	0.366/0.141	0.505/0.222	0.415/0.165	0.456/0.173	0.263/0.107	0.328/0.119	0.401/0.153	0.461/0.194
RAMC [126]	**0.460**/**0.244**	**0.388**/**0.251**	**0.735**/**0.395**	**0.644**/**0.357**	**0.658**/**0.374**	**0.740**/**0.393**	**0.604**/**0.343**	**0.349**/**0.190**	**0.589**/**0.302**	**0.634**/**0.327**	**0.707**/**0.384**
STAR (Ours)	**0.656**/**0.433**	**0.476**/**0.312**	**0.729**/**0.488**	**0.685**/**0.448**	**0.781**/**0.521**	**0.706**/**0.467**	**0.738**/**0.487**	**0.673**/**0.433**	**0.717**/**0.450**	**0.695**/**0.461**	**0.774**/**0.523**

There are three outputs: a probability map indicating object presence, an offset map for center correction, and a scale map for size determination. These outputs are optimized through a multitask learning framework, where the probability map provides coarse localization, while the offset and scale maps enhance precision. The optimization employs focal loss (L_{cls}) for classification [110], complemented by L1 loss (L_1) and generalized IoU loss (L_{giou}) for regression tasks [111]. The composite loss function (L_{total}) integrates these components for model training

$$L_{\text{total}} = L_{\text{cls}} + \varphi_{L_1} L_1 + \varphi_{\text{giou}} L_{\text{giou}} \tag{12}$$

where φ_{L_1} and φ_{giou} are regularization coefficients assigned values of 5.0 and 2.0, respectively.

IV. EXPERIMENT AND DISCUSSION

A. Experimental Setup

1) Experimental Overview: Evaluations cover three popular SVOT benchmarks: SatSOT [4], SV248S [49], and OOTB [3]. Our STAR undergoes comparative analysis against more than 50 SOTA trackers, systematically categorized as: i) GV methods employing handcrafted features (e.g., CSK [25], CN [112], DAT [113], SRDCF [56], Staple [28], BACF [54], CSRDCF [57], and STRCF [27]), ii) Siamese and Transformer-based GV solutions (e.g., SiamRPN [36], DaSiamRPN [62], SiamRPN++ [69], UpdateNet [101], SiamDW [70], SiamMask [114], SiamBAN [115], SiamFC++ [116], SiamGAT [117], LightTrack [118], TransT [72], Stark [39], SiamCAR [119], OSTrack [76], SimTrack [120], SBT [78], TCTrack [121], GRM [122], SeqTrack [100], TCTrack++ [98], ARTrack [123], HiT [80], AVTrack [83], ODTrack [104], SMAT [124], AQATrack [105], EVPTrack [106], HIPTrack [125], ARTrackV2 [77], SGLATrack [84], and ORTrack [85]), and iii) SVOT methods (e.g., CFME [20], DF [22], RAMC [126], CPKF [23], SiamMDM [15], SiamTM [14], SVLPNet [41], REPS [33], DOCPF [24], MACF [6], TATrans [7], and SiamS2F [8]). Ablation analysis confirms critical design choices. Notably, consistent weights and configurations are maintained throughout benchmark verification, ensuring fair comparisons.

2) Implementation Detail: The foundation model undergoes 150-epoch pretraining on GOT-10k [12], TrackingNet [13], LaSOT [11], and COCO [10], optimized by AdamW [127] with a weight decay of 10^{-4} applied at the 120th epoch. The backbone employs HiViT-Base [128], with an initial learning rate of 4×10^{-5} for the backbone and 4×10^{-4} for all other parameters. Training uses common data augmentations, including brightness jittering and horizontal flip. Subsequent 60-epoch fine-tuning on SatSOT-train [42] applies progressive learning rate decay from 4×10^{-4} to 4×10^{-5} at the 48th epoch. The SatSOT-train subset (947 sequences: 703 cars, 65 planes, 149 ships, and 30 trains) inherits from SatMTB [129], including distinctive spatial resolutions, motion patterns, and object scale distributions. Our implementation configures the search region of $256 \times 256 \times 3$ and template of $128 \times 128 \times 3$ with video-level sampling (batch size: 40), developed in PyTorch 2.0.0 on two RTX 4090 GPUs. Table I presents the details of the tracking model, including the backbone, resolution, tunable memory parameters (TMPs), total memory parameters (Params), floating point operations (FLOPs), and frames per second (FPS).

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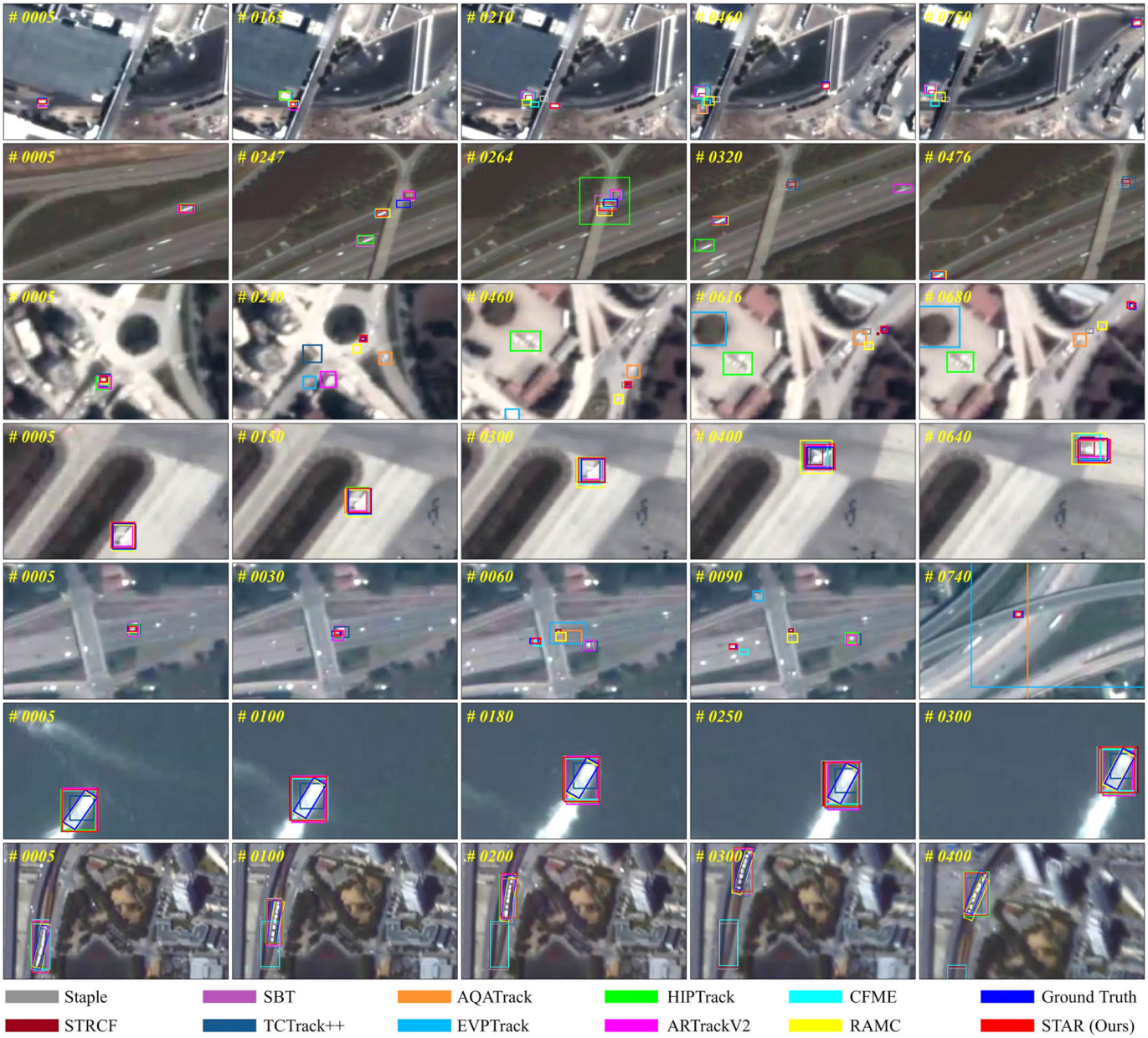


Fig. 12. Visual analysis, arranged from top to bottom: car_01 (SatSOT), car_61 (SatSOT), 03_000036 (SV248S), 04_000003 (SV248S), 05_000035 (SV248S), ship_10 (OOTB), and train_1 (OOTB).

3) Assessment Criteria: Tracking performance is quantified by one-pass evaluation using precision and success plots [130]. The precision plot quantifies frame proportions where center location errors remain below specified pixel thresholds (1–50). Conversely, the success plot calculates temporal segments where overlap scores surpass predefined thresholds (0–1). Algorithm ranking follows two criteria: the 5-pixel precision metric and the area under the curve (AUC) of the success plot, corresponding to precision rate (PR) and success rate (SR), respectively. In particular, the OOTB [3] employs official validation protocols incorporating PR, normalized PR (NPR), and SR for comprehensive analysis. Inference speed is reported as FPS. Superior tracking capability correlates with higher PR, NPR, SR, and FPS.

B. Performance Comparison on SatSOT

1) SatSOT Overview: SatSOT [4] features 105 test scenarios with 27 664 frames containing frame-level horizontal bounding box (HBB) annotations for SVOT. It categorizes primary objects into four classes: 65 cars, 26 trains, nine planes, and five ships, reflecting diverse spatial scales and significant challenges. SV clips demonstrate variable durations averaging 263 frames (range: 120–750 frames). Sourced from three commercial SV platforms (Skybox, Jilin-1, and Carbonite-2), it incorporates 11 distinct attributes for systematic tracker evaluation: illumination variation (IV), BC, background jitter (BJT), deformation (DEF), POC, FOC, rotation (ROT), similar object (SOB), low quality (LQ), aspect ratio change (ARC), and tiny object (TO).

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TABLE VI
PERFORMANCE ASSESSMENT ON OOTB

No.	Trackers	Venues	PR	NPR	SR
1	CSK [25]	*ECCV 2012*	0.540	0.578	0.478
2	CN [112]	*CVPR 2014*	0.671	0.686	0.554
3	DAT [113]	*CVPR 2015*	0.474	0.569	0.453
4	SRDCF [56]	*ICCV 2015*	0.766	0.744	0.584
5	Staple [28]	*CVPR 2016*	0.772	0.738	0.607
6	BACF [54]	*ICCV 2017*	0.795	0.750	0.547
7	CSRDCF [57]	*IJCV 2018*	0.788	0.768	**0.655**
8	STRCF [27]	*CVPR 2018*	0.795	0.756	0.584
9	SiamRPN [36]	*CVPR 2018*	0.605	0.624	0.464
10	DaSiamRPN [62]	*ECCV 2018*	0.578	0.646	0.515
11	SiamRPN++ [69]	*CVPR 2019*	0.758	0.735	0.589
12	UpdateNet [101]	*ICCV 2019*	0.652	0.708	0.550
13	SiamDW [70]	*CVPR 2019*	0.800	0.764	**0.644**
14	SiamMask [114]	*CVPR 2019*	0.747	0.729	0.552
15	SiamBAN [115]	*CVPR 2020*	0.754	0.733	0.523
16	SiamFC++ [116]	*AAAI 2020*	0.668	0.672	0.530
17	SiamGAT [117]	*CVPR 2021*	0.692	0.686	0.474
18	LightTrack [118]	*CVPR 2021*	0.611	0.616	0.487
19	TransT [72]	*CVPR 2021*	0.711	0.698	0.550
20	Stark [39]	*ICCV 2021*	0.610	0.601	0.477
21	SiamCAR [119]	*IJCV 2022*	**0.824**	0.779	0.607
22	OSTrack [76]	*ECCV 2022*	0.665	0.650	0.515
23	SimTrack [120]	*ECCV 2022*	0.535	0.544	0.430
24	SBT [78]	*CVPR 2022*	0.700	0.682	0.525
25	TCTrack [121]	*CVPR 2022*	0.681	0.659	0.484
26	GRM [122]	*CVPR 2023*	0.645	0.629	0.500
27	SeqTrack [100]	*CVPR 2023*	0.726	0.720	0.550
28	TCTrack++ [98]	*TPAMI 2023*	0.753	0.722	0.504
29	ARTrack [123]	*CVPR 2023*	0.762	0.731	0.549
30	HiT [80]	*ICCV 2023*	0.553	0.557	0.426
31	AVTrack [83]	*ICML 2024*	0.647	0.665	0.519
32	ODTrack [104]	*AAAI 2024*	0.686	0.695	0.526
33	SMAT [124]	*WACV 2024*	0.688	0.665	0.526
34	AQATrack [105]	*CVPR 2024*	0.773	0.741	0.579
35	EVPTrack [106]	*AAAI 2024*	0.741	0.722	0.562
36	HIPTrack [125]	*CVPR 2024*	0.773	0.737	0.585
37	ARTrackV2 [77]	*CVPR 2024*	**0.823**	**0.782**	0.598
38	SGLATrack [84]	*CVPR 2025*	0.709	0.699	0.542
39	ORTrack [85]	*CVPR 2025*	0.668	0.666	0.524
40	CFME [20]	*TGRS 2020*	0.753	0.714	0.610
41	DF [22]	*JSTARS 2022*	0.758	0.747	0.623
42	RAMC [126]	*RS 2022*	0.781	0.703	0.598
43	CPKF [23]	*TGRS 2022*	0.796	0.772	0.512
44	REPS [33]	*JAG 2024*	0.689	0.664	0.454
45	DOCPF [24]	*TGRS 2024*	0.804	0.776	0.514
46	TATrans [7]	*TGRS 2024*	0.664	0.695	0.518
47	SiamS2F [8]	*TGRS 2025*	0.807	**0.795**	0.486
48	STAR	*Ours*	**0.846**	**0.826**	**0.678**

2) Comparative Analysis With GV Trackers: Comparative analysis is conducted between STAR and leading GV trackers utilizing handcrafted and deep features. As detailed in Table II and Fig. 8, STAR demonstrates superior performance across evaluation criteria. STAR attains leading PR/SR metrics (0.639/0.537), outperforming top ARTrackV2 by 1.3% PR and 5.9% SR margins. In particular, STAR yields 9.4% and 9.6% higher SR margins compared to leading ARTrack and SiamCAR, respectively. Attributed to its superior spatiotemporal modeling capability for small-sized satellite objects, STAR achieves robust tracking performance.

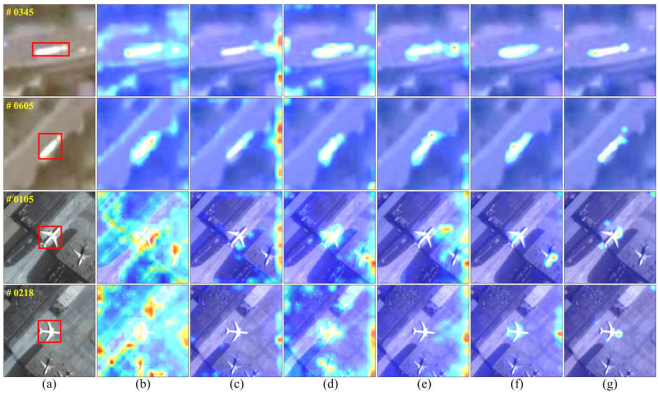


Fig. 13. Visualization of attention maps and its evolutionary process. (a) Ground truth. (b)–(f) Attention maps of the ECAM for unified relation modeling layers 1, 5, 10, 15, and 20, respectively. (g) Attention map of the TDM (from the temporal decoding layer).

Specifically, the SEM enhances low-contrast satellite scenes, enabling robust discrimination through improved multimodal representations. The unified fusion of TDM's deep temporal attention and INM's physics-aware navigation mitigates occlusion and reduces tracking drift. Detailed results are presented in Table II. Comprehensive evaluations confirm STAR's superiority across multiple performance metrics.

3) Comparative Analysis With SV Trackers: Table III and Fig. 8 outline the characteristics and outcomes of the compared SVOT arts. STAR outperforms the leading MACF and DOCPF by 4.2% and 5.6% in SR and by 0.2% and 2.8% in PR, respectively. The superior performance of STAR is primarily due to the designed hierarchical Transformer architecture in ECAM, which addresses modality limitations with dual correlation layers. The local relation modeling layer ensures precise localization, while the unified relation modeling layer with SMRA enables cross-modal alignment. In addition, the INM reduces occlusion-induced drift by modeling tracking confidence-uncertainty, outperforming conventional SV trackers. These innovations, along with SEM's dynamic range optimization for low-contrast scenes, demonstrate STAR's ability in spatiotemporal fusion.

4) Attribute-Wise Performance Analysis: Quantitative evaluations across individual attributes are detailed in Fig. 9. Regarding the SR measurement, our solution consistently maintains leading positions across nearly all 11 challenging attributes, culminating in optimal overall (OVE) SR performance. FOC presents particular challenges for trackers that rely only on appearance features. Through synergistic integration of spatial and temporal patterns of objects, STAR effectively addresses this limitation and achieves the SR of 0.564, yielding 19.5% improvement over SBT [78]. Notably, the TDM integrates historical object states for end-to-end reasoning, incorporating deep temporal features. In addition, the INM models physical temporal features, including an awareness selector to assess the confidence-uncertainty and an inertial navigation scheme to manage anomalous interferences (e.g., FOC). Experimental outcomes confirm STAR's capability in SVOT scenarios.

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TABLE VII
CATEGORY-WISE AND ATTRIBUTE-WISE NPR EVALUATIONS ON OOTB

Trackers	Category-Wise Evaluations				Attribute-Wise Evaluations											
	Car	Ship	Train	Plane	DEF	IPR	PO	FO	IV	MB	BC	OON	SA	LT	IM	AM
CSK [25]	0.471	0.554	0.507	0.827	0.628	0.562	0.444	0.387	0.568	0.575	0.547	0.583	0.447	0.395	0.493	0.370
CN [112]	0.627	0.656	0.596	0.865	0.698	0.687	0.586	0.401	0.670	0.672	0.674	0.634	0.597	0.552	0.555	0.552
DAT [113]	0.472	0.660	0.380	0.711	0.555	0.501	0.525	0.395	0.596	0.584	0.537	0.508	0.426	0.424	0.468	0.445
BACF [54]	**0.750**	**0.800**	0.327	0.861	0.585	0.638	**0.769**	**0.542**	**0.747**	**0.778**	**0.774**	0.584	**0.659**	**0.700**	**0.786**	**0.736**
SiamRPN [36]	0.544	0.597	0.464	0.862	0.672	0.624	0.509	0.347	0.618	0.613	0.586	0.528	0.517	0.447	0.579	0.437
UpdateNet [101]	**0.752**	0.532	0.624	0.875	0.714	**0.728**	**0.709**	0.509	0.676	0.654	0.680	0.716	**0.653**	0.571	0.632	**0.758**
SiamMask [114]	0.739	0.773	0.297	0.830	0.579	0.703	0.650	0.515	0.700	0.708	**0.733**	0.586	0.636	**0.665**	0.760	0.717
SiamBAN [115]	0.700	0.752	0.394	0.905	0.631	**0.753**	0.668	0.475	0.724	0.746	0.714	0.579	0.643	0.603	0.697	0.715
SiamFC++ [116]	0.650	0.625	0.286	**0.921**	0.519	0.678	0.644	0.531	0.651	0.688	0.661	0.579	0.599	0.476	0.695	0.636
SiamGAT [117]	0.693	0.691	0.155	0.883	0.440	0.681	0.591	0.459	0.692	0.728	0.686	0.477	0.613	0.545	**0.777**	0.651
LightTrack [118]	0.545	0.596	0.305	0.891	0.549	0.587	0.489	0.434	0.615	0.602	0.592	0.523	0.515	0.431	0.657	0.493
Stark [39]	0.432	0.765	0.356	0.804	0.473	0.529	0.428	0.406	0.621	0.629	0.593	0.532	0.407	0.468	0.485	0.436
OSTrack [76]	0.518	0.598	**0.726**	**0.921**	0.777	0.642	0.558	0.502	0.636	0.662	0.624	**0.778**	0.582	0.482	0.634	0.567
SimTrack [120]	0.330	0.593	0.580	0.857	0.668	0.505	0.388	0.373	0.588	0.569	0.508	0.615	0.422	0.366	0.420	0.341
SBT [78]	0.489	**0.843**	0.533	0.895	0.635	0.586	0.466	**0.538**	0.723	**0.770**	0.681	0.659	0.546	0.575	0.589	0.516
TCTrack [121]	0.590	0.614	0.428	**0.932**	0.587	0.653	0.624	0.509	0.692	0.703	0.635	0.570	0.552	0.527	0.577	0.627
GRM [122]	0.448	0.640	0.682	**0.921**	0.764	0.599	0.536	0.386	0.650	0.671	0.592	0.734	0.517	0.467	0.555	0.486
TCTrack++ [98]	0.607	0.799	0.470	**0.936**	0.674	0.685	0.600	0.525	**0.741**	0.767	0.704	0.628	0.600	0.593	0.710	0.564
SMAT [124]	0.531	0.793	0.335	0.883	0.601	0.620	0.608	0.479	0.679	0.709	0.635	0.536	0.507	0.538	0.574	0.516
EVPTrack [106]	0.593	0.770	0.665	0.920	**0.783**	0.705	0.620	0.535	0.728	0.760	0.703	0.736	0.622	0.602	0.667	0.652
RAMC [126]	0.624	0.663	**0.768**	0.865	**0.821**	0.707	0.640	0.388	0.706	0.659	0.681	**0.739**	0.628	0.574	0.714	0.593
STAR (Ours)	**0.763**	**0.877**	**0.813**	0.886	**0.871**	**0.791**	**0.756**	**0.715**	**0.829**	**0.820**	**0.825**	**0.876**	**0.752**	**0.784**	**0.779**	**0.734**

TABLE VIII
CATEGORY-WISE AND ATTRIBUTE-WISE SR EVALUATIONS ON OOTB

Trackers	Category-Wise Evaluations				Attribute-Wise Evaluations											
	Car	Ship	Train	Plane	DEF	IPR	PO	FO	IV	MB	BC	OON	SA	LT	IM	AM
CSK [25]	0.359	0.459	0.412	0.740	0.522	0.443	0.355	0.310	0.478	0.477	0.451	0.467	0.365	0.319	0.379	0.288
CN [112]	0.474	0.521	0.485	**0.765**	0.563	0.533	0.469	0.314	0.546	0.548	0.540	0.498	0.481	0.429	0.410	0.426
DAT [113]	0.376	0.505	0.292	0.594	0.446	0.385	0.419	0.310	0.475	0.465	0.428	0.408	0.340	0.326	0.366	0.358
BACF [54]	0.499	0.612	0.249	0.673	0.445	0.450	**0.538**	**0.438**	0.540	0.568	**0.561**	0.422	0.454	**0.479**	0.541	0.469
SiamRPN [36]	0.442	0.460	0.337	0.560	0.524	0.460	0.417	0.282	0.446	0.443	0.438	0.410	0.407	0.351	0.459	0.363
UpdateNet [101]	**0.600**	0.452	0.440	0.623	0.545	**0.562**	**0.585**	0.412	0.517	0.502	0.531	0.543	**0.510**	0.455	0.491	**0.607**
SiamMask [114]	**0.581**	0.610	0.188	0.575	0.437	0.524	0.513	0.399	0.516	0.541	0.553	0.439	0.487	**0.506**	**0.589**	**0.555**
SiamBAN [115]	0.532	0.581	0.206	0.562	0.424	0.534	0.510	0.335	0.504	0.533	0.516	0.398	0.469	0.459	0.530	0.536
SiamFC++ [116]	0.517	0.505	0.186	0.720	0.404	0.530	0.495	0.402	0.503	0.538	0.525	0.435	0.473	0.377	0.564	0.502
SiamGAT [117]	0.495	0.531	0.093	0.521	0.312	0.468	0.437	0.351	0.456	0.495	0.484	0.349	0.427	0.398	0.565	0.445
LightTrack [118]	0.422	0.499	0.172	0.715	0.400	0.444	0.380	0.344	0.483	0.480	0.468	0.386	0.394	0.324	0.525	0.366
Stark [39]	0.337	0.613	0.274	0.647	0.381	0.409	0.326	0.306	0.490	0.496	0.474	0.422	0.324	0.366	0.386	0.345
OSTrack [76]	0.402	0.494	**0.602**	0.706	**0.643**	0.498	0.438	0.389	0.503	0.520	0.497	**0.636**	0.456	0.377	0.508	0.438
SimTrack [120]	0.265	0.484	0.476	0.645	0.545	0.393	0.310	0.306	0.458	0.452	0.404	0.510	0.336	0.290	0.339	0.278
SBT [78]	0.373	**0.662**	0.413	0.677	0.503	0.448	0.357	0.415	0.553	**0.582**	0.524	0.520	0.409	0.430	0.456	0.391
TCTrack [121]	0.456	0.498	0.235	0.616	0.397	0.468	0.460	0.363	0.504	0.526	0.478	0.404	0.406	0.413	0.464	0.477
GRM [122]	0.355	0.524	0.571	0.704	0.634	0.471	0.426	0.318	0.512	0.527	0.475	0.609	0.414	0.371	0.454	0.383
TCTrack++ [98]	0.461	0.575	0.231	0.603	0.426	0.464	0.438	0.369	0.515	0.546	0.501	0.423	0.419	0.430	0.536	0.425
SMAT [124]	0.417	**0.628**	0.244	0.711	0.475	0.482	0.467	0.371	0.532	0.561	0.505	0.428	0.400	0.411	0.469	0.413
EVPTrack [106]	0.463	0.607	0.543	0.696	0.641	0.544	0.484	**0.417**	**0.558**	**0.592**	0.549	0.599	0.479	0.460	0.532	0.509
RAMC [126]	0.507	0.564	**0.636**	**0.788**	**0.694**	**0.601**	0.526	0.319	**0.603**	0.563	**0.575**	**0.621**	**0.524**	0.466	**0.582**	0.483
STAR (Ours)	**0.598**	**0.719**	**0.646**	**0.783**	**0.723**	**0.639**	**0.593**	**0.586**	**0.685**	**0.666**	**0.669**	**0.699**	**0.599**	**0.611**	**0.620**	**0.577**

C. Performance Comparison on SV248S

1) SV248S Overview: SV248S [49] (https://github.com/xdai-dlgvv/SV248S) contains 248 annotated SV objects comprising 156 621 frames captured at the original pixel resolution of 2160 × 4096. Derived from Jilin-1 SV recorded at 25.0 FPS, the sequences span 500–753 frames (e.g., 20–30 s

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TABLE IX
DETAILS AND COMPOSITION OF ABLATION VARIANTS ON SATSOT AND SV248S

Models	SEM	ECAM	INM	SatSOT		SV248S		Overall	
				PR	SR	PR	SR	PR	SR
Model-1	-	-	-	0.533	0.417	0.465	0.200	0.499	0.309
Model-2	✓	-	-	0.579	0.464	0.601	0.341	0.590	0.403
Model-3	✓	✓	-	**0.607**	**0.517**	**0.665**	**0.455**	**0.636**	**0.486**
Model-4	✓	-	✓	**0.599**	**0.480**	**0.677**	**0.378**	**0.638**	**0.429**
STAR (Ours)	✓	✓	✓	**0.639**	**0.537**	**0.774**	**0.523**	**0.707**	**0.530**

TABLE X
ABLATION STUDIES ON INSERTING SMRA LAYER INDEXES WITHIN ECAM

Models	SEM	ECAM	INM	L-Index	SatSOT		SV248S		Overall	
					PR	SR	PR	SR	PR	SR
Model-1	-	-	-	-	0.533	0.417	0.465	0.200	0.499	0.309
Model-5	✓	✓	✓	[5]	**0.618**	**0.526**	**0.771**	**0.517**	**0.695**	**0.522**
Model-6	✓	✓	✓	[5, 10]	**0.612**	0.512	**0.754**	**0.508**	**0.683**	**0.510**
Model-7	✓	✓	-	[5]	0.600	**0.514**	0.690	0.468	0.645	0.491
Model-8	✓	✓	-	[5, 10]	0.588	0.499	0.671	0.458	0.630	0.479
Model-3	✓	✓	-	[5, 10, 15]	0.607	**0.517**	0.665	0.455	0.636	0.486
Model-9	✓	✓	-	[0: 19]	0.583	0.488	0.658	0.450	0.621	0.469
Model-10	✓	✓	✓	[0: 19]	**0.608**	0.505	**0.728**	**0.492**	0.668	0.499
STAR (Ours)	✓	✓	✓	[5, 10, 15]	**0.639**	**0.537**	**0.774**	**0.523**	**0.707**	**0.530**

TABLE XI
ABLATION STUDIES OF THE TDM

Models	TDM	SatSOT		SV248S		Overall	
		PR	SR	PR	SR	PR	SR
Model-1	-	**0.533**	**0.417**	**0.465**	**0.200**	**0.499**	**0.309**
Model-11	-	**0.633**	**0.535**	**0.764**	**0.516**	**0.699**	**0.526**
STAR (Ours)	✓	**0.639**	**0.537**	**0.774**	**0.523**	**0.707**	**0.530**

duration) across four nations (United States, Bahrain, Libya, and Türkiye) and five urban areas (San Francisco, Minneapolis, Muharraq, Derna, and Adana). Four primary object classes are defined: vehicle, large vehicle (L-vehicle), airplane, and ship. Ten SV-specific attributes are systematically categorized: natural disturbance (ND), short-term occlusion (STO), long-term occlusion (LTO), DS, background change (BCH), IV, slow motion (SM), continuous occlusion (CO), background cluster (BCL), and in-plane ROT (IPR). Each sequence is classified into three difficulty levels: simple, normal, and hard. For standardized evaluation, all objects are annotated with polygon masks converted to HBB representations.

2) Comparative Analysis With GV Trackers: The proposed method is evaluated against SOTA GV methods. As summarized in Table II and Fig. 10, STAR attains a PR/SR score of 0.774/0.523, securing the top position. The Staple tracker relies on correlation filters and combines various handcrafted features to achieve competitive performance among compared methods. However, on the one hand, traditional handcrafted features and correlation filters may limit tracking performance. In contrast, deep appearance features, commonly used in SVOT methods, consist of a shallow layer that encodes low-level information with high spatial resolution for precise localization, and a deep layer that captures high-level semantic information, invariant to certain object changes. Our method, on the other hand, designs a multimodal Transformer architecture with SMRA to facilitate feature extraction, relation learning, and domain adaptation, achieving competitive results. It exhibits a gain of 2.4%/4.2% in PR/SR over Staple, validating marked results.

3) Comparative Analysis With SV Trackers: Performance metrics for mainstream SVOT approaches are listed in Table II and Fig. 10. Recent methods, such as SiamS2F, design a Siamese spectral–spatial-frame correlation network through feature refinement. However, it struggles to predict the object's location due to a lack of motion simulation and interference-aware capabilities. In contrast, the proposed method combines deep and physical temporal features to enable continuous trajectory modeling and address anomalies like DS and LTO, resulting in superior performance. STAR surpasses SiamS2F, SiamTM, SVLPNet, and SiamMDM by 1.7%, 2.8%, 8.6%, and 6.3% in PR and 1.7%, 3.6%, 5.2%, and 7.0% in SR, respectively. These comparisons highlight the effectiveness of STAR in SVOT.

4) Category-Wise Performance Analysis: We further analyze performance across four object categories on SV248S: vehicle, L-vehicle, airplane, and ship, as illustrated in Table IV. The proposed method ranks among the top three performers in PR/SR for three categories (vehicle, L-vehicle, and ship). Competitors show tighter curves and higher PR and SR scores in the airplane category, suggesting its relative

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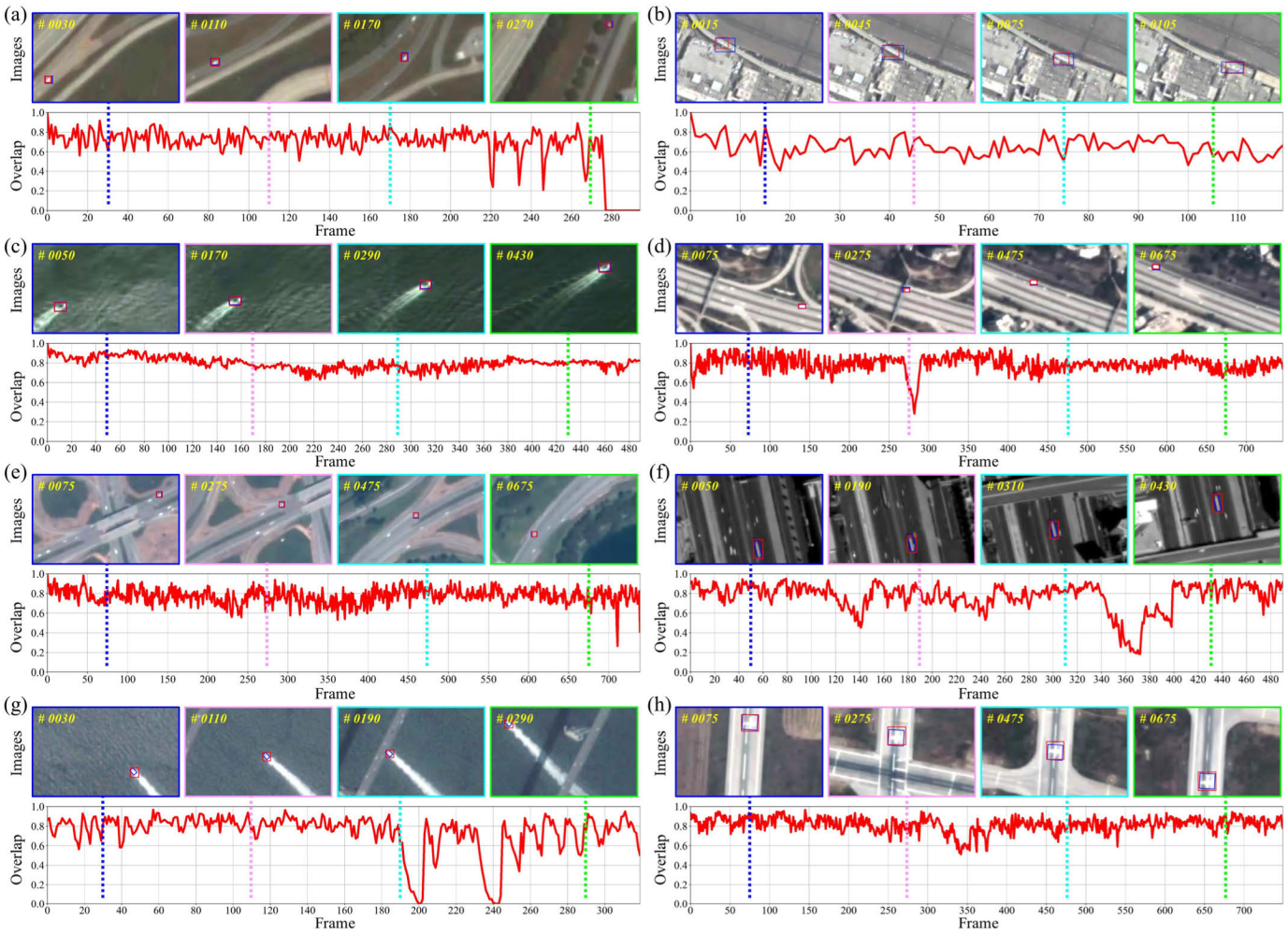


Fig. 14. Overlap curves and tracking samples of STAR in diverse scenarios: (a) car_12 (SatSOT), (b) train_20 (SatSOT), (c) 01_000029 (SV248S), (d) 03_000008 (SV248S), (e) 05_000013 (SV248S), (f) car_21 (OOTB), (g) ship_14 (OOTB), and (h) plane_4 (OOTB). Predicted results are depicted by red bounding boxes, with ground truth annotations shown in blue. This comprehensive analysis validates STAR's adaptability.

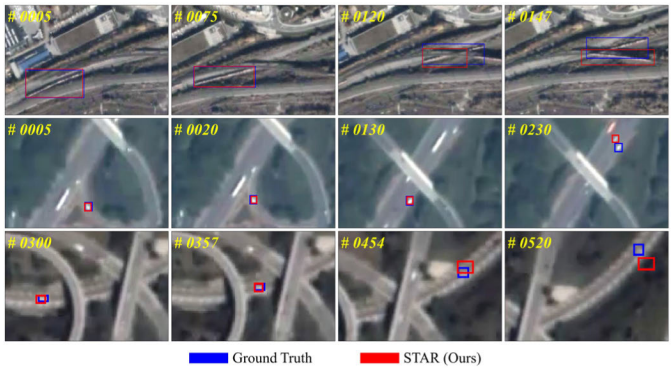


Fig. 15. Visualization of failure scenarios, arranged from top to bottom: train_25 (SatSOT), 04_000018 (SV248S), and 05_000008 (SV248S).

simplicity. In this case, the proposed method also produces competitive results with PR and SR of 0.864 and 0.765, respectively. Conversely, most trackers exhibit suboptimal performance in the vehicle category, emphasizing the need for enhanced small-object tracking solutions. STAR achieves leading PR/SR scores of 0.751/0.490 in this challenging category, underscoring its robustness for SV-based small-object applications.

5) Difficulty-Wise Performance Analysis: Table IV presents a performance analysis across tracking difficulties for various methods on SV248S. Regarding precision metrics, STAR demonstrates exceptional performance, achieving top rankings in two (normal and hard) out of three difficulty categories, while maintaining competitive standing in the simple category, ultimately securing the OVE premier position. In SR evaluations, STAR consistently ranks within the top three across all difficulty levels.

6) Attribute-Wise Performance Analysis: As detailed in Table V, STAR attains the top three rankings across all ten attributes on the SV248S benchmark, achieving the highest OVE performance. Its modular architecture enables robust handling of complex SVOT scenarios, establishing a baseline for tracking applications.

D. Performance Comparison on OOTB

1) OOTB Overview: OOTB [3] comprises 29 890 frames from 110 SV sequences spanning four object categories: car, ship, train, and plane. Each sequence is annotated with oriented bounding boxes and labeled with 12 attributes: DEF, IPR, POC, FOC, IV, motion blur (MB), BCs, out-of-normal (OON), similar appearance (SA), less textures (LTs), isotropic

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TABLE XII
SENSITIVITY ANALYSIS OF THE TRACKING CUM THRESHOLD

No.	CUM	SatSOT		SV248S		Overall	
		PR	SR	PR	SR	PR	SR
1	0	0.607	0.517	0.665	0.455	0.636	0.486
2	0.05	0.607	0.518	0.672	0.459	0.640	0.489
3	0.10	0.620	0.522	0.690	0.471	0.655	0.497
4	0.15	0.622	0.526	0.704	0.480	0.663	0.503
5	0.20	0.636	0.535	0.714	0.485	0.675	0.510
6	0.25	0.636	0.536	0.727	0.494	0.682	0.515
7	0.30	0.636	0.534	0.740	0.501	0.688	0.518
8	0.35	0.635	0.532	0.760	0.513	0.698	0.523
9	0.40	0.639	0.537	0.774	0.523	0.707	0.530
10	0.45	0.621	0.523	0.790	0.532	0.706	0.528
11	0.50	0.622	0.519	0.791	0.533	0.707	0.526
12	0.55	0.599	0.501	0.796	0.533	0.698	0.517
13	0.60	0.575	0.484	0.793	0.531	0.684	0.508
14	0.65	0.551	0.461	0.777	0.517	0.664	0.489
15	0.70	0.535	0.451	0.746	0.496	0.641	0.474
16	0.75	0.474	0.409	0.692	0.453	0.583	0.431
17	0.80	0.417	0.377	0.613	0.398	0.515	0.388
18	0.85	0.373	0.343	0.505	0.323	0.439	0.333
19	0.90	0.326	0.312	0.396	0.247	0.361	0.280
20	0.95	0.298	0.296	0.318	0.198	0.308	0.247
21	1.0	0.286	0.282	0.255	0.158	0.271	0.220

CUM means the tracking confidence-uncertainty metric. For enhanced visualization, column values are color-graded from blue (minimum) to red (maximum).

motion (IM), and anisotropic motion (AM). The benchmark (https://github.com/YZCU/OOTB) adopts a rigorous evaluation protocol to ensure fairness and comprehensiveness.

2) Comparative Analysis With GV and SV Trackers: STAR is evaluated against SOTA GV and SV trackers. Table VI and Fig. 11 summarize the results. With PR/NPR/SR scores of 0.846/0.826/0.678, STAR surpasses SOTA trackers, demonstrating superior robustness.

3) Category-Wise Performance Analysis: OOTB's four object categories are analyzed. Tables VII and VIII summarize the main NPR and SR results. STAR secures leading NPR positions in three categories (car, ship, and train) and ranks within the top two for SR in all four object categories.

4) Attribute-Wise Performance Analysis: Tracking robustness is systematically evaluated across individual attributes. As documented in Tables VII and VIII, STAR secures the top three positions in both NPR and SR metrics for all 12 attributes.

E. Visual Presentation

1) Qualitative Analysis: Fig. 12 presents qualitative comparisons of multiple trackers under challenging scenarios. Trackers like Staple, SBT, AQATrack, HIPTrack, and CFME demonstrate early tracking drift under rotational and full occlusion scenarios, as evidenced in car_01 (SatSOT). In contrast, the proposed STAR can maintain a continuous trajectory throughout the sequence. When processing very low-resolution SV sequences, e.g., 03_000036 (SV248S), several SOTA methods exhibit compromised performance characterized by either complete failure or inaccurate scale estimation, primarily due to insufficient feature discriminability. Furthermore, trackers like EVPTrack and ARTrackV2 show limited robustness against STO, DS, and BCH, as shown in 05_000035 (SV248S), particularly under CO. By integrating spatial patterns with deep-physical temporal representations, our STAR achieves superior cross-category and cross-attribute adaptability.

2) Attention Map Analysis: To further highlight the efficacy of ECAM and TDM in STAR, the attention maps, which are generated by the unified relation modeling layer in ECAM and the temporal decoding layer in TDM, are visualized as shown in Fig. 13. These visualizations demonstrate that the tracking network, directed by spatial and temporal prompts, effectively concentrates on the object's position. The results confirm that the network can utilize these multidimensional prompts to accurately focus on the object throughout the tracking process.

F. Ablation Study and Sensitivity Analysis

1) Ablation Overview: STAR integrates four key components: SEM, ECAM, TDM, and INM. To evaluate their contributions, ablation studies are conducted on SatSOT and SV248S. In addition, sensitivity analysis is performed on the CUM thresholds to optimize configurations. Comprehensive results are presented in Tables IX–XII.

2) Ablation Study for SEM: SEM enhances SV scenes by generating a brightening auxiliary modality to amplify weak features. To quantify its impact, ablation experiments are performed, as detailed in Table IX. Model-1 serves as the baseline, while Model-2 introduces SEM. Compared to Model-1, Model-2 achieves significant improvements in PR/SR by 4.6%/4.7% and 13.6%/14.1% on the SatSOT and SV248S benchmarks, respectively. OVE, Model-2 outperforms Model-1 by 9.1% in PR and 9.4% in SR, demonstrating the critical role of SEM.

3) Ablation Study for ECAM: ECAM comprises a multimodal Transformer architecture with SMRA to facilitate feature extraction, relation learning, and domain adaptation. Ablation studies are conducted to assess SMRA's contribution, as summarized in Table IX. Model-3, which includes SMRA, achieves a 4.6%/8.3% improvement in OVE PR/SR over Model-2. Compared to Model-4, STAR exhibits substantial gains of 6.9%/10.1% in OVE PR/SR, respectively. To further optimize ECAM, SMRA is embedded into the unified relation modeling layers, generating diverse embedding fashions. Table X outlines the experimental setups and results, where the L-Index denotes the SMRA layer position within the unified relation modeling layers. Dense ablation variants on the L-Index are constructed with and without INM, revealing that our model consistently achieves superior performance with appropriate additional parameters. These findings highlight the potential for further performance enhancements within ECAM.

4) Ablation Study for TDM: TDM enhances temporal state transitions by integrating deep temporal features and spatial features. It includes a temporal decoding layer and a spatiotemporal fusion layer to capture historical state changes.

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Ablation experiments, detailed in Table XI, evaluate TDM's contribution. Model-11, which excludes TDM from our STAR, demonstrates the module's significance. With deep temporal prompts, STAR achieves improvements of 0.8% and 0.4% in OVE PR and SR, respectively. Fig. 14 provides overlap curves and tracking samples, validating the effectiveness of spatiotemporal fusion of the proposed method.

5) Ablation Study for INM: INM models physical temporal features, including an awareness selector to assess the tracking confidence-uncertainty, and an inertial navigation scheme to manage anomalous interferences. As shown in Table IX, a comparison between Model-2 and Model-4 reveals enhancements of 4.8% and 2.6% in OVE PR and SR due to INM inclusion. Similarly, removing INM from STAR results in reductions of 7.1% and 4.4% in PR and SR, as observed in the comparison between Model-3 and STAR. Consistent trends are evident in other model pairs (Model-5 and Model-7, Model-6 and Model-8, and Model-9 and Model-10), as detailed in Table X, further confirming INM's effectiveness.

6) Sensitivity Analysis for CUM Threshold: We explore the sensitivity of the CUM threshold (τ). Table XII presents exhaustive experiments with τ ranging from 0 to 1.0, where values are color-coded from blue (low) to red (high). Results indicate that performance initially improves and then declines as τ increases, with an optimal result observed at $\tau = 0.40$.

OVE, these systematic evaluations highlight the contributions of major components to STAR's performance, while the sensitivity analysis provides insights into optimal configurations for SVOT.

G. Limitation and Discussion

Despite its OVE robustness, STAR encounters challenges in complex scenarios. As illustrated in Fig. 15, in the train_25 (SatSOT) sequence, the train has a small aspect ratio. When subjected to occlusion, similar interference, and extreme IVs, the proposed method experiences tracking drifts. Specifically, in severe and prolonged full occlusion scenarios, the method may lose track of small objects, as demonstrated by the sequence of 05_000008 (SV248S). Here, a small object is heavily occluded by bridges for an extended period and changes its trajectory, leading to temporal tracking failure. The above cases highlight limitations in handling LTO and abrupt motion. In addition, SV objects often exhibit complex motion patterns, such as acceleration and sudden turns. Quadratic regression may be insufficient for modeling such dynamics. To improve trajectory continuity, future work should explore higher order motion models and adaptive filtering techniques (e.g., particle filters). Integrating physics-informed neural networks may also enable end-to-end learning of motion patterns, enhancing generalization across diverse scenarios.

V. Conclusion

This study proposes STAR, a unified spatiotemporal fusion framework for SVOT. By introducing a SEM, STAR optimizes SV scenes, generating refined multimodal representations. It also devises the ECAM, which comprises a multimodal hierarchical Transformer architecture with local and unified relation modeling to jointly achieve feature extraction, relation learning, and domain adaptation. The TDM integrates historical object states, enhancing spatial features with deep temporal information. Moreover, the INM models physical temporal features, incorporating an awareness selector to evaluate the tracking confidence-uncertainty and an inertial navigation scheme to reduce anomalous interferences, ensuring continuous trajectory. With minimal tunable parameters, STAR outperforms counterparts on multiple SVOT benchmarks, providing a robust solution for remote sensing visual tasks.

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